

AWS Certified Solutions Architect - Associate SAA-C03

question 0401-0500

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【題號】question 401

【題目】

A company wants to use the AWS Cloud to make an existing application highly available and resilient. The current version of the application resides in the company's data center. The application recently experienced data loss after a database server crashed because of an unexpected power outage.

The company needs a solution that avoids any single points of failure. The solution must give the application the ability to scale to meet user demand.

Which solution will meet these requirements?

一家公司希望使用 AWS 雲端使現有應用程式具有高可用性和彈性。該應用程式的目前版本駐留在公司的資料中心。最近，由於意外斷電，資料庫伺服器崩潰，該應用程式出現了資料遺失的情況。

該公司需要一種能夠避免任何單點故障的解決方案。該解決方案必須使應用程式能夠擴展以滿足用戶需求。

哪種解決方案可以滿足這些要求？

【選項】

A. Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Use an Amazon RDS DB instance in a Multi-AZ configuration.

B. Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group in a single Availability Zone. Deploy the database on an EC2 instance. Enable EC2 Auto Recovery.

C. Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Use an Amazon RDS DB instance with a read replica in a

single Availability Zone. Promote the read replica to replace the primary DB instance if the primary DB instance fails.

D. Deploy the application servers by using Amazon EC2 instances in an Auto Scaling group across multiple Availability Zones. Deploy the primary and secondary database servers on EC2 instances across multiple Availability Zones. Use Amazon Elastic Block Store (Amazon EBS) Multi-Attach to create shared storage between the instances.

【參考答案】A

【題號】question 402

【題目】

A company needs to ingest and handle large amounts of streaming data that its application generates. The application runs on Amazon EC2 instances and sends data to Amazon Kinesis Data Streams, which is configured with default settings. Every other day, the application consumes the data and writes the data to an Amazon S3 bucket for business intelligence (BI) processing. The company observes that Amazon S3 is not receiving all the data that the application sends to Kinesis Data Streams.

What should a solutions architect do to resolve this issue?

公司需要攝取和處理其應用程式產生的大量串流資料。該應用程式在 Amazon EC2 執行個體上執行，並將資料傳送至使用預設設定配置的 Amazon Kinesis Data Streams。每隔一天，應用程式就會使用資料並將資料寫入 Amazon S3 儲存桶以進行商業智慧 (BI) 處理。該公司觀察到，Amazon S3 未接收應用程式發送到 Kinesis Data Streams 的所有資料。

解決方案架構師應該做什麼來解決這個問題？

【選項】

A. Update the Kinesis Data Streams default settings by modifying the data retention period.

B. Update the application to use the Kinesis Producer Library (KPL) to send the data to Kinesis Data Streams.

C. Update the number of Kinesis shards to handle the throughput of the data that is sent to Kinesis Data Streams.

D.Turn on S3 Versioning within the S3 bucket to preserve every version of every object that is ingested in the S3 bucket.

【參考答案】A

【題號】question 403

【題目】

A developer has an application that uses an AWS Lambda function to upload files to Amazon S3 and needs the required permissions to perform the task. The developer already has an IAM user with valid IAM credentials required for Amazon S3.

What should a solutions architect do to grant the permissions?

開發人員擁有一個使用 AWS Lambda 函數將檔案上傳到 Amazon S3 的應用程式，並且需要執行該任務所需的權限。開發人員已經擁有一個 IAM 用戶，該用戶具有 Amazon S3 所需的有效 IAM 憑證。

解決方案架構師應該做什麼來授予權限？

【選項】

- A.Add required IAM permissions in the resource policy of the Lambda function.
- B.Create a signed request using the existing IAM credentials in the Lambda function.
- C.Create a new IAM user and use the existing IAM credentials in the Lambda function.
- D.Create an IAM execution role with the required permissions and attach the IAM role to the Lambda function.

【參考答案】D

【題號】question 404

【題目】

A company has deployed a serverless application that invokes an AWS Lambda function when new documents are uploaded to an Amazon S3 bucket. The application uses the Lambda function to process the documents. After a recent marketing campaign, the company noticed that the application did not process many of the documents.

What should a solutions architect do to improve the architecture of this application?

一家公司部署了一個無伺服器應用程式，當新文件上傳到 Amazon S3 儲存桶時，該應用程式會呼叫 AWS Lambda 函數。該應用程式使用 Lambda 函數來處理文件。經過最近的營銷活動後，該公司注意到該應用程式沒有處理許多文件。

解決方案架構師應該如何改進該應用程式的架構？

【選項】

- A. Set the Lambda function's runtime timeout value to 15 minutes.
- B. Configure an S3 bucket replication policy. Stage the documents in the S3 bucket for later processing.
- C. Deploy an additional Lambda function. Load balance the processing of the documents across the two Lambda functions.
- D. Create an Amazon Simple Queue Service (Amazon SQS) queue. Send the requests to the queue. Configure the queue as an event source for Lambda.

【參考答案】D

【題號】question 405

【題目】

A solutions architect is designing the architecture for a software demonstration environment. The environment will run on Amazon EC2 instances in an Auto Scaling group behind an Application Load Balancer (ALB). The system will experience significant increases in traffic during working hours but is not required to operate on weekends.

Which combination of actions should the solutions architect take to ensure that the system can scale to meet demand? (Choose two.)

解決方案架構師正在設計軟體來示範環境的架構。此環境將在應用程式負載平衡器 (ALB) 後面的 Auto Scaling 群組中的 Amazon EC2 執行個體上執行。該系統在工作時間內的流量將顯著增加, 但不需要在周末運行。

解決方案架構師應採取哪種操作組合來確保系統可以擴展以滿足需求? (選兩個。)

【選項】

- A. Use AWS Auto Scaling to adjust the ALB capacity based on request rate.
- B. Use AWS Auto Scaling to scale the capacity of the VPC internet gateway.
- C. Launch the EC2 instances in multiple AWS Regions to distribute the load across Regions.
- D. Use a target tracking scaling policy to scale the Auto Scaling group based on instance CPU utilization.
- E. Use scheduled scaling to change the Auto Scaling group minimum, maximum, and desired capacity to zero for weekends. Revert to the default values at the start of the week.

【參考答案】DE

【題號】question 406

【題目】

A solutions architect is designing a two-tiered architecture that includes a public subnet and a database subnet. The web servers in the public subnet must be open to the internet on port 443. The Amazon RDS for MySQL DB instance in the database subnet must be accessible only to the web servers on port 3306.

Which combination of steps should the solutions architect take to meet these requirements? (Choose two.)

解決方案架構師正在設計一個兩層架構，其中包括公共子網路 and 資料庫子網路。公有子網路中的 Web 伺服器必須透過連接埠 443 對網際網路開放。資料庫子網路中的 Amazon RDS for MySQL 資料庫執行個體必須只能由連接埠 3306 上的 Web 伺服器存取。

解決方案架構師應該採取哪些步驟組合來滿足這些要求？（選兩個。）

【選項】

A.Create a network ACL for the public subnet. Add a rule to deny outbound traffic to 0.0.0.0/0 on port 3306.

B.Create a security group for the DB instance. Add a rule to allow traffic from the public subnet CIDR block on port 3306.

C.Create a security group for the web servers in the public subnet. Add a rule to allow traffic from 0.0.0.0/0 on port 443.

D.Create a security group for the DB instance. Add a rule to allow traffic from the web servers' security group on port 3306.

E.Create a security group for the DB instance. Add a rule to deny all traffic except traffic from the web servers' security group on port 3306.

【參考答案】CD

【題號】question 407

【題目】

A company is implementing a shared storage solution for a gaming application that is hosted in the AWS Cloud. The company needs the ability to use Lustre clients to access data. The solution must be fully managed.

Which solution meets these requirements?

一家公司正在為 AWS 雲端中託管的遊戲應用程式實施共享儲存解決方案。該公司需要能夠使用 Lustre 客戶端存取資料。該解決方案必須經過全面管理。

哪種解決方案滿足這些要求？

【選項】

A.Create an AWS DataSync task that shares the data as a mountable file system. Mount the file system to the application server.

B.Create an AWS Storage Gateway file gateway. Create a file share that uses the required client protocol. Connect the application server to the file share.

C.Create an Amazon Elastic File System (Amazon EFS) file system, and configure it to support Lustre. Attach the file system to the origin server. Connect the application server to the file system.

D.Create an Amazon FSx for Lustre file system. Attach the file system to the origin server. Connect the application server to the file system.

【參考答案】D

【題號】question 408

【題目】

A company runs an application that receives data from thousands of geographically dispersed remote devices that use UDP. The application processes the data immediately and sends a message back to the device if necessary. No data is stored.

The company needs a solution that minimizes latency for the data transmission from the devices. The solution also must provide rapid failover to another AWS Region.

Which solution will meet these requirements?

一家公司運行一個應用程序，該應用程式從數千個使用 UDP 的地理分散的遠端設備接收資料。應用程式立即處理數據，並在必要時將訊息傳回裝置。不儲存任何資料。

該公司需要一種能夠最大限度地減少設備資料傳輸延遲的解決方案。該解決方案還必須提供到另一個 AWS 區域的快速故障轉移。

哪種解決方案可以滿足這些要求？

【選項】

A. Configure an Amazon Route 53 failover routing policy. Create a Network Load Balancer (NLB) in each of the two Regions. Configure the NLB to invoke an AWS Lambda function to process the data.

B. Use AWS Global Accelerator. Create a Network Load Balancer (NLB) in each of the two Regions as an endpoint. Create an Amazon Elastic Container Service (Amazon ECS) cluster with the Fargate launch type. Create an ECS service on the cluster. Set the ECS service as the target for the NLB. Process the data in Amazon ECS.

C. Use AWS Global Accelerator. Create an Application Load Balancer (ALB) in each of the two Regions as an endpoint. Create an Amazon Elastic Container Service (Amazon ECS) cluster with the Fargate launch type. Create an ECS service on the cluster. Set the ECS service as the target for the ALB. Process the data in Amazon ECS.

D. Configure an Amazon Route 53 failover routing policy. Create an Application Load Balancer (ALB) in each of the two Regions. Create an Amazon Elastic Container Service (Amazon ECS) cluster with the Fargate launch type. Create an ECS service on the cluster. Set the ECS service as the target for the ALB. Process the data in Amazon ECS.

【參考答案】B

【題號】question 409

【題目】

A solutions architect must migrate a Windows Internet Information Services (IIS) web application to AWS. The application currently relies on a file share hosted in the user's on-premises network-attached storage (NAS). The solutions architect has proposed migrating the IIS web servers to Amazon EC2 instances in multiple Availability Zones that are connected to the storage solution, and configuring an Elastic Load Balancer attached to the instances.

Which replacement to the on-premises file share is MOST resilient and durable?

解決方案架構師必須將 Windows Internet Information Services (IIS) Web 應用程式移轉到 AWS。該應用程式目前依賴用戶本地網路附加儲存 (NAS) 中託管的檔案共用。解決方案架構師建議將 IIS Web 伺服器移轉到連接到儲存解決方案的多個可用區中的 Amazon EC2 實例，並配置附加到執行個體的彈性負載平衡器。

哪種本地文件共享替代方案最具彈性和耐用性？

【選項】

- A.Migrate the file share to Amazon RDS.
- B.Migrate the file share to AWS Storage Gateway.
- C.Migrate the file share to Amazon FSx for Windows File Server.
- D.Migrate the file share to Amazon Elastic File System (Amazon EFS).

【參考答案】C

【題號】question 410

【題目】

A company is deploying a new application on Amazon EC2 instances. The application writes data to Amazon Elastic Block Store (Amazon EBS) volumes. The company needs to ensure that all data that is written to the EBS volumes is encrypted at rest.

Which solution will meet this requirement?

一家公司正在 Amazon EC2 執行個體上部署新應用程式。該應用程式將資料寫入 Amazon Elastic Block Store (Amazon EBS) 磁碟區。公司需要確保寫入 EBS 磁碟區的所有資料都經過靜態加密。

哪種解決方案可以滿足這個要求？

【選項】

- A.Create an IAM role that specifies EBS encryption. Attach the role to the EC2 instances.
- B.Create the EBS volumes as encrypted volumes. Attach the EBS volumes to the EC2 instances.
- C.Create an EC2 instance tag that has a key of Encrypt and a value of True. Tag all instances that require encryption at the EBS level.
- D.Create an AWS Key Management Service (AWS KMS) key policy that enforces EBS encryption in the account. Ensure that the key policy is active.

【參考答案】B

【題號】question 411

【題目】

A company has a web application with sporadic usage patterns. There is heavy usage at the beginning of each month, moderate usage at the start of each week, and unpredictable usage during the week. The application consists of a web server and a MySQL database server running inside the data center. The company would like to move the application to the AWS Cloud, and needs to select a cost-effective database platform that will not require database modifications.

Which solution will meet these requirements?

一家公司有一個具有零星使用模式的 Web 應用程式。每個月初使用量較大，每週開始使用量中等，週內使用量無法預測。該應用程式由在資料中心內運行的 Web 伺服器 and MySQL 資料庫伺服器組成。該公司希望將應用程式遷移到AWS雲，並且需要選擇一個經濟高效且不需要修改資料庫的資料庫平台。

哪種解決方案可以滿足這些要求？

【選項】

- A.Amazon DynamoDB
- B.Amazon RDS for MySQL
- C.MySQL-compatible Amazon Aurora Serverless
- D.MySQL deployed on Amazon EC2 in an Auto Scaling group

【參考答案】C

【題號】question 412

【題目】

An image-hosting company stores its objects in Amazon S3 buckets. The company wants to avoid accidental exposure of the objects in the S3 buckets to the public. All S3 objects in the entire AWS account need to remain private.

Which solution will meet these requirements?

一家圖像託管公司將其物件儲存在 Amazon S3 儲存桶中。該公司希望避免 S3 儲存桶中的物件意外暴露給公眾。整個 AWS 帳戶中的所有 S3 物件都需要保持私有。

哪種解決方案可以滿足這些要求？

【選項】

A. Use Amazon GuardDuty to monitor S3 bucket policies. Create an automatic remediation action rule that uses an AWS Lambda function to remediate any change that makes the objects public.

B. Use AWS Trusted Advisor to find publicly accessible S3 buckets. Configure email notifications in Trusted Advisor when a change is detected. Manually change the S3 bucket policy if it allows public access.

C. Use AWS Resource Access Manager to find publicly accessible S3 buckets. Use Amazon Simple Notification Service (Amazon SNS) to invoke an AWS Lambda function when a change is detected. Deploy a Lambda function that programmatically remediates the change.

D. Use the S3 Block Public Access feature on the account level. Use AWS Organizations to create a service control policy (SCP) that prevents IAM users from changing the setting. Apply the SCP to the account.

【參考答案】D

【題號】question 413

【題目】

An ecommerce company is experiencing an increase in user traffic. The company's store is deployed on Amazon EC2 instances as a two-tier web application consisting of a web tier and a separate database tier. As traffic increases, the company notices that the architecture is causing significant delays in sending timely marketing and order confirmation email to users. The company wants to reduce the time it spends resolving complex email delivery issues and minimize operational overhead.

What should a solutions architect do to meet these requirements?

一家電子商務公司的用戶流量正在增加。該公司的商店作為兩層 Web 應用程式部署在 Amazon EC2 執行個體上，由 Web 層和單獨的資料庫層組成。隨著流量的增加，該公司注意到該架構導致向用戶發送及時行銷和訂單確認電子郵件方面出現嚴重延遲。該公司希望減少解決複雜電子郵件傳送問題所花費的時間並最大限度地減少營運開銷。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

- A. Create a separate application tier using EC2 instances dedicated to email processing.
- B. Configure the web instance to send email through Amazon Simple Email Service (Amazon SES).
- C. Configure the web instance to send email through Amazon Simple Notification Service (Amazon SNS).
- D. Create a separate application tier using EC2 instances dedicated to email processing. Place the instances in an Auto Scaling group.

【參考答案】B

【題號】question 414

【題目】

A company has a business system that generates hundreds of reports each day. The business system saves the reports to a network share in CSV format. The company needs to store this data in the AWS Cloud in near-real time for analysis.

Which solution will meet these requirements with the LEAST administrative overhead?

一家公司有一個每天產生數百份報告的業務系統。業務系統將報告以 CSV 格式儲存至網路共用。該公司需要近乎即時地將這些資料儲存在 AWS 雲端中以供分析。

哪種解決方案能夠以最少的管理開銷來滿足這些要求？

【選項】

A. Use AWS DataSync to transfer the files to Amazon S3. Create a scheduled task that runs at the end of each day.

B. Create an Amazon S3 File Gateway. Update the business system to use a new network share from the S3 File Gateway.

C. Use AWS DataSync to transfer the files to Amazon S3. Create an application that uses the DataSync API in the automation workflow.

D. Deploy an AWS Transfer for SFTP endpoint. Create a script that checks for new files on the network share and uploads the new files by using SFTP.

【參考答案】B

【題號】question 415

【題目】

A company is storing petabytes of data in Amazon S3 Standard. The data is stored in multiple S3 buckets and is accessed with varying frequency. The company does not know access patterns for all the data. The company needs to implement a solution for each S3 bucket to optimize the cost of S3 usage.

Which solution will meet these requirements with the MOST operational efficiency?

一家公司正在 Amazon S3 Standard 中儲存 PB 級的資料。資料儲存在多個 S3 儲存桶中，並以不同的頻率存取。該公司不知道所有數據的存取模式。該公司需要為每個 S3 儲存桶實施解決方案，以優化 S3 使用成本。

哪種解決方案能夠以最高的營運效率滿足這些要求？

【選項】

- A.Create an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 Intelligent-Tiering.
- B.Use the S3 storage class analysis tool to determine the correct tier for each object in the S3 bucket. Move each object to the identified storage tier.
- C.Create an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 Glacier Instant Retrieval.
- D.Create an S3 Lifecycle configuration with a rule to transition the objects in the S3 bucket to S3 One Zone-Infrequent Access (S3 One Zone-IA).

【參考答案】A

【題號】question 416

【題目】

A rapidly growing global ecommerce company is hosting its web application on AWS. The web application includes static content and dynamic content. The website stores online transaction processing (OLTP) data in an Amazon RDS database. The website's users are experiencing slow page loads.

Which combination of actions should a solutions architect take to resolve this issue? (Choose two.)

一家快速發展的全球電子商務公司正在 AWS 上託管其 Web 應用程式。Web 應用程式包括靜態內容和動態內容。該網站將線上事務處理 (OLTP) 資料儲存在 Amazon RDS 資料庫中。該網站的使用者遇到頁面載入緩慢的問題。

解決方案架構師應採取哪種操作組合來解決此問題？（選兩個。）

【選項】

- A.Configure an Amazon Redshift cluster.
- B.Set up an Amazon CloudFront distribution.
- C.Host the dynamic web content in Amazon S3.
- D.Create a read replica for the RDS DB instance.

E.Configure a Multi-AZ deployment for the RDS DB instance.

【參考答案】BD

【題號】question 417

【題目】

A company uses Amazon EC2 instances and AWS Lambda functions to run its application. The company has VPCs with public subnets and private subnets in its AWS account. The EC2 instances run in a private subnet in one of the VPCs. The Lambda functions need direct network access to the EC2 instances for the application to work.

The application will run for at least 1 year. The company expects the number of Lambda functions that the application uses to increase during that time. The company wants to maximize its savings on all application resources and to keep network latency between the services low.

Which solution will meet these requirements?

一家公司使用 Amazon EC2 執行個體和 AWS Lambda 函數來執行其應用程式。該公司的 AWS 帳戶中擁有帶有公有子網路和私人子網路的 VPC。EC2 執行個體在其中一個 VPC 的私有子網路中運作。Lambda 函數需要直接網路存取 EC2 執行個體才能讓應用程式正常運作。

該應用程式將運行至少 1 年。該公司預計在此期間應用程式使用的 Lambda 函數數量將會增加。該公司希望最大限度地節省所有應用程式資源，並保持服務之間的網路延遲較低。

哪種解決方案可以滿足這些要求？

【選項】

A.Purchase an EC2 Instance Savings Plan Optimize the Lambda functions' duration and memory usage and the number of invocations. Connect the Lambda functions to the private subnet that contains the EC2 instances.

B.Purchase an EC2 Instance Savings Plan Optimize the Lambda functions' duration and memory usage, the number of invocations, and the amount of data that is transferred. Connect the Lambda functions to a public subnet in the same VPC where the EC2 instances run.

C. Purchase a Compute Savings Plan. Optimize the Lambda functions' duration and memory usage, the number of invocations, and the amount of data that is transferred. Connect the Lambda functions to the private subnet that contains the EC2 instances.

D. Purchase a Compute Savings Plan. Optimize the Lambda functions' duration and memory usage, the number of invocations, and the amount of data that is transferred. Keep the Lambda functions in the Lambda service VPC.

【參考答案】C

【題號】question 418

【題目】

A solutions architect needs to allow team members to access Amazon S3 buckets in two different AWS accounts: a development account and a production account. The team currently has access to S3 buckets in the development account by using unique IAM users that are assigned to an IAM group that has appropriate permissions in the account.

The solutions architect has created an IAM role in the production account. The role has a policy that grants access to an S3 bucket in the production account.

Which solution will meet these requirements while complying with the principle of least privilege?

解決方案架構師需要允許團隊成員存取兩個不同 AWS 帳戶中的 Amazon S3 儲存桶：開發帳戶和生產帳戶。該團隊目前可以使用分配給在帳戶中具有適當權限的 IAM 群組的唯一 IAM 使用者來存取開發帳戶中的 S3 儲存桶。

解決方案架構師已在生產帳戶中建立 IAM 角色。該角色具有授予對生產帳戶中的 S3 儲存桶的存取權限的策略。

哪一種解決方案能夠滿足這些要求，同時遵守最小特權原則？

【選項】

A. Attach the Administrator Access policy to the development account users.

B. Add the development account as a principal in the trust policy of the role in the production account.

- C. Turn off the S3 Block Public Access feature on the S3 bucket in the production account.
- D. Create a user in the production account with unique credentials for each team member.

【參考答案】B

【題號】question 419

【題目】

A company uses AWS Organizations with all features enabled and runs multiple Amazon EC2 workloads in the ap-southeast-2 Region. The company has a service control policy (SCP) that prevents any resources from being created in any other Region. A security policy requires the company to encrypt all data at rest.

An audit discovers that employees have created Amazon Elastic Block Store (Amazon EBS) volumes for EC2 instances without encrypting the volumes. The company wants any new EC2 instances that any IAM user or root user launches in ap-southeast-2 to use encrypted EBS volumes. The company wants a solution that will have minimal effect on employees who create EBS volumes.

Which combination of steps will meet these requirements? (Choose two.)

一家公司使用啟用了所有功能的 AWS Organizations，並在 ap-southeast-2 區域運行多個 Amazon EC2 工作負載。該公司有一項服務控制策略 (SCP)，可防止在任何其他區域建立任何資源。安全策略要求公司對所有靜態資料進行加密。

審計發現員工為 EC2 執行個體建立了 Amazon Elastic Block Store (Amazon EBS) 卷，但未加密這些磁碟區。該公司希望任何 IAM 用戶或根用戶在 ap-southeast-2 中啟動的任何新 EC2 執行個體都使用加密的 EBS 磁碟區。該公司希望找到對創建 EBS 磁碟區的員工影響最小的解決方案。

哪一種步驟組合可以滿足這些要求？（選兩個。）

【選項】

A. In the Amazon EC2 console, select the EBS encryption account attribute and define a default encryption key.

B.Create an IAM permission boundary. Attach the permission boundary to the root organizational unit (OU). Define the boundary to deny the ec2:CreateVolume action when the ec2:Encrypted condition equals false.

C.Create an SCP. Attach the SCP to the root organizational unit (OU). Define the SCP to deny the ec2:CreateVolume action when the ec2:Encrypted condition equals false.

D.Update the IAM policies for each account to deny the ec2:CreateVolume action when the ec2:Encrypted condition equals false.

E.In the Organizations management account, specify the Default EBS volume encryption setting.

【參考答案】CE

【題號】question 420

【題目】

A company wants to use an Amazon RDS for PostgreSQL DB cluster to simplify time-consuming database administrative tasks for production database workloads. The company wants to ensure that its database is highly available and will provide automatic failover support in most scenarios in less than 40 seconds. The company wants to offload reads off of the primary instance and keep costs as low as possible.

Which solution will meet these requirements?

一家公司希望使用 Amazon RDS for PostgreSQL 資料庫叢集來簡化生產資料庫工作負載的耗時資料庫管理任務。該公司希望確保其資料庫具有高可用性，並將在大多數場景下在 40 秒內提供自動故障轉移支援。該公司希望卸載主實例的讀取任務並盡可能降低成本。

哪種解決方案可以滿足這些要求？

【選項】

A.Use an Amazon RDS Multi-AZ DB instance deployment. Create one read replica and point the read workload to the read replica.

B.Use an Amazon RDS Multi-AZ DB duster deployment Create two read replicas and point the read workload to the read replicas.

C. Use an Amazon RDS Multi-AZ DB instance deployment. Point the read workload to the secondary instances in the Multi-AZ pair.

D. Use an Amazon RDS Multi-AZ DB cluster deployment. Point the read workload to the reader endpoint.

【參考答案】D

【題號】question 421

【題目】

A company runs a highly available SFTP service. The SFTP service uses two Amazon EC2 Linux instances that run with elastic IP addresses to accept traffic from trusted IP sources on the internet. The SFTP service is backed by shared storage that is attached to the instances. User accounts are created and managed as Linux users in the SFTP servers.

The company wants a serverless option that provides high IOPS performance and highly configurable security. The company also wants to maintain control over user permissions.

Which solution will meet these requirements?

一家公司運行高度可用的 SFTP 服務。SFTP 服務使用兩個使用彈性 IP 位址運作的 Amazon EC2 Linux 執行個體來接受來自網際網路上受信任 IP 來源的流量。SFTP 服務由附加到執行個體的共用儲存提供支援。使用者帳戶在 SFTP 伺服器中以 Linux 使用者身分建立和管理。

該公司想要一種能夠提供高 IOPS 效能和高度可設定安全性的無伺服器選項。該公司還希望保持對用戶權限的控制。

哪種解決方案可以滿足這些要求？

【選項】

A. Create an encrypted Amazon Elastic Block Store (Amazon EBS) volume. Create an AWS Transfer Family SFTP service with a public endpoint that allows only trusted IP addresses. Attach the EBS volume to the SFTP service endpoint. Grant users access to the SFTP service.

B. Create an encrypted Amazon Elastic File System (Amazon EFS) volume. Create an AWS Transfer Family SFTP service with elastic IP addresses and a VPC endpoint that has

internet-facing access. Attach a security group to the endpoint that allows only trusted IP addresses. Attach the EFS volume to the SFTP service endpoint. Grant users access to the SFTP service.

C.Create an Amazon S3 bucket with default encryption enabled. Create an AWS Transfer Family SFTP service with a public endpoint that allows only trusted IP addresses. Attach the S3 bucket to the SFTP service endpoint. Grant users access to the SFTP service.

D.Create an Amazon S3 bucket with default encryption enabled. Create an AWS Transfer Family SFTP service with a VPC endpoint that has internal access in a private subnet. Attach a security group that allows only trusted IP addresses. Attach the S3 bucket to the SFTP service endpoint. Grant users access to the SFTP service.

【參考答案】B

【題號】question 422

【題目】

A company is developing a new machine learning (ML) model solution on AWS. The models are developed as independent microservices that fetch approximately 1 GB of model data from Amazon S3 at startup and load the data into memory. Users access the models through an asynchronous API. Users can send a request or a batch of requests and specify where the results should be sent.

The company provides models to hundreds of users. The usage patterns for the models are irregular. Some models could be unused for days or weeks. Other models could receive batches of thousands of requests at a time.

Which design should a solutions architect recommend to meet these requirements?

一家公司正在 AWS 上開發新的機器學習 (ML) 模型解決方案。這些模型被開發為獨立的微服務，在啟動時從 Amazon S3 獲取大約 1 GB 的模型資料並將資料載入記憶體。使用者透過非同步 API 存取模型。用戶可以發送一個請求或一批請求，並指定結果應發送到何處。

該公司為數百名用戶提供模型。模型的使用模式是不規則的。某些型號可能會閒置數天或數週。其他模型可以一次接收數千個批次的請求。

解決方案架構師應該推薦哪一種設計來滿足這些要求？

【選項】

- A. Direct the requests from the API to a Network Load Balancer (NLB). Deploy the models as AWS Lambda functions that are invoked by the NLB.
- B. Direct the requests from the API to an Application Load Balancer (ALB). Deploy the models as Amazon Elastic Container Service (Amazon ECS) services that read from an Amazon Simple Queue Service (Amazon SQS) queue. Use AWS App Mesh to scale the instances of the ECS cluster based on the SQS queue size.
- C. Direct the requests from the API into an Amazon Simple Queue Service (Amazon SQS) queue. Deploy the models as AWS Lambda functions that are invoked by SQS events. Use AWS Auto Scaling to increase the number of vCPUs for the Lambda functions based on the SQS queue size.
- D. Direct the requests from the API into an Amazon Simple Queue Service (Amazon SQS) queue. Deploy the models as Amazon Elastic Container Service (Amazon ECS) services that read from the queue. Enable AWS Auto Scaling on Amazon ECS for both the cluster and copies of the service based on the queue size.

【參考答案】D

【題號】question 423

【題目】

A solutions architect wants to use the following JSON text as an identity-based policy to grant specific permissions:

Which IAM principals can the solutions architect attach this policy to? (Choose two.)

解決方案架構師希望使用以下 JSON 文字作為基於身分的策略來授予特定權限：

解決方案架構師可以將此策略附加到哪些 IAM 主體？（選兩個。）

```
{  "Statement": [
    {
      "Action": [
        "ssm:ListDocuments",
        "ssm:GetDocument"
      ],
      "Effect": "Allow",
      "Resource": "*",
      "Sid": ""
    }
  ],
  "Version": "2012-10-17"
}
```

【選項】

- A.Role
- B.Group
- C.Organization
- D.Amazon Elastic Container Service (Amazon ECS) resource
- E.Amazon EC2 resource

【參考答案】AB

【題號】question 424

【題目】

A company is running a custom application on Amazon EC2 On-Demand Instances. The application has frontend nodes that need to run 24 hours a day, 7 days a week and backend nodes that need to run only for a short time based on workload. The number of backend nodes varies during the day.

The company needs to scale out and scale in more instances based on workload.

Which solution will meet these requirements MOST cost-effectively?

一家公司正在 Amazon EC2 隨選執行個體上執行自訂應用程式。應用程式的前端節點需要每週 7 天、每天 24 小時運行，而後端節點則根據工作負載僅需要運行很短的時間。後端節點的數量在一天中會有所不同。

公司需要根據工作負載橫向擴展和擴展更多實例。

哪一種解決方案能夠最具成本效益地滿足這些要求？

【選項】

A. Use Reserved Instances for the frontend nodes. Use AWS Fargate for the backend nodes.

B. Use Reserved Instances for the frontend nodes. Use Spot Instances for the backend nodes.

C. Use Spot Instances for the frontend nodes. Use Reserved Instances for the backend nodes.

D. Use Spot Instances for the frontend nodes. Use AWS Fargate for the backend nodes.

【參考答案】B

【題號】question 425

【題目】

A company uses high block storage capacity to runs its workloads on premises. The company's daily peak input and output transactions per second are not more than 15,000 IOPS. The company wants to migrate the workloads to Amazon EC2 and to provision disk performance independent of storage capacity.

Which Amazon Elastic Block Store (Amazon EBS) volume type will meet these requirements MOST cost-effectively?

一家公司使用高區塊儲存容量在本地運行其工作負載。公司每天每秒的峰值輸入輸出交易不超過15,000 IOPS。該公司希望將工作負載遷移到 Amazon EC2，並獨立於儲存容量配置磁碟效能。

哪種 Amazon Elastic Block Store (Amazon EBS) 磁碟區類型能夠最經濟高效地滿足這些要求？

【選項】

- A.GP2 volume type
- B.io2 volume type
- C.GP3 volume type
- D.io1 volume type

【參考答案】C

【題號】question 426

【題目】

A company needs to store data from its healthcare application. The application's data frequently changes. A new regulation requires audit access at all levels of the stored data.

The company hosts the application on an on-premises infrastructure that is running out of storage capacity. A solutions architect must securely migrate the existing data to AWS while satisfying the new regulation.

Which solution will meet these requirements?

公司需要儲存來自其醫療保健應用程式的資料。應用程式的數據經常更改。新法規要求對所有級別的儲存資料進行審計存取。

該公司將應用程式託管在儲存容量不足的本地基礎設施上。解決方案架構師必須將現有資料安全地遷移到 AWS，同時滿足新法規。

哪種解決方案可以滿足這些要求？

【選項】

- A.Use AWS DataSync to move the existing data to Amazon S3. Use AWS CloudTrail to log data events.
- B.Use AWS Snowcone to move the existing data to Amazon S3. Use AWS CloudTrail to log management events.

C. Use Amazon S3 Transfer Acceleration to move the existing data to Amazon S3. Use AWS CloudTrail to log data events.

D. Use AWS Storage Gateway to move the existing data to Amazon S3. Use AWS CloudTrail to log management events.

【參考答案】A

【題號】question 427

【題目】

A solutions architect is implementing a complex Java application with a MySQL database. The Java application must be deployed on Apache Tomcat and must be highly available.

What should the solutions architect do to meet these requirements?

解決方案架構師正在使用 MySQL 資料庫實作複雜的 Java 應用程式。Java 應用程式必須部署在 Apache Tomcat 上，並且必須具有高可用性。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

A. Deploy the application in AWS Lambda. Configure an Amazon API Gateway API to connect with the Lambda functions.

B. Deploy the application by using AWS Elastic Beanstalk. Configure a load-balanced environment and a rolling deployment policy.

C. Migrate the database to Amazon ElastiCache. Configure the ElastiCache security group to allow access from the application.

D. Launch an Amazon EC2 instance. Install a MySQL server on the EC2 instance. Configure the application on the server. Create an AMI. Use the AMI to create a launch template with an Auto Scaling group.

【參考答案】B

【題號】question 428

【題目】

A serverless application uses Amazon API Gateway, AWS Lambda, and Amazon DynamoDB. The Lambda function needs permissions to read and write to the DynamoDB table.

Which solution will give the Lambda function access to the DynamoDB table MOST securely?

無伺服器應用程式使用 Amazon API Gateway、AWS Lambda 和 Amazon DynamoDB。Lambda 函數需要讀取和寫入 DynamoDB 表的權限。

哪種解決方案可以讓 Lambda 函數最安全地存取 DynamoDB 表？

【選項】

A.Create an IAM user with programmatic access to the Lambda function. Attach a policy to the user that allows read and write access to the DynamoDB table. Store the access_key_id and secret_access_key parameters as part of the Lambda environment variables. Ensure that other AWS users do not have read and write access to the Lambda function configuration.

B.Create an IAM role that includes Lambda as a trusted service. Attach a policy to the role that allows read and write access to the DynamoDB table. Update the configuration of the Lambda function to use the new role as the execution role.

C.Create an IAM user with programmatic access to the Lambda function. Attach a policy to the user that allows read and write access to the DynamoDB table. Store the access_key_id and secret_access_key parameters in AWS Systems Manager Parameter Store as secure string parameters. Update the Lambda function code to retrieve the secure string parameters before connecting to the DynamoDB table.

D.Create an IAM role that includes DynamoDB as a trusted service. Attach a policy to the role that allows read and write access from the Lambda function. Update the code of the Lambda function to attach to the new role as an execution role.

【參考答案】B

【題號】question 429

【題目】

The following IAM policy is attached to an IAM group. This is the only policy applied to the group.

What are the effective IAM permissions of this policy for group members?

以下 IAM 策略附加到 IAM 群組。這是應用於該組的唯一策略。

該策略對群組成員的有效 IAM 權限有哪些？

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "1",
      "Effect": "Allow",
      "Action": "ec2:*",
      "Resource": "*",
      "Condition": {
        "StringEquals": {
          "ec2:Region": "us-east-1"
        }
      }
    },
    {
      "Sid": "2",
      "Effect": "Deny",
      "Action": [
        "ec2:StopInstances",
        "ec2:TerminateInstances"
      ],
      "Resource": "*",
      "Condition": {
        "BoolIfExists": {"aws:MultiFactorAuthPresent": false}
      }
    }
  ]
}
```

【選項】

A.Group members are permitted any Amazon EC2 action within the us-east-1 Region. Statements after the Allow permission are not applied.

B.Group members are denied any Amazon EC2 permissions in the us-east-1 Region unless they are logged in with multi-factor authentication (MFA).

C.Group members are allowed the ec2:StopInstances and ec2:TerminateInstances permissions for all Regions when logged in with multi-factor authentication (MFA). Group members are permitted any other Amazon EC2 action.

D.Group members are allowed the ec2:StopInstances and ec2:TerminateInstances permissions for the us-east-1 Region only when logged in with multi-factor authentication (MFA). Group members are permitted any other Amazon EC2 action within the us-east-1 Region.

【參考答案】D

【題號】question 430

【題目】

A manufacturing company has machine sensors that upload .csv files to an Amazon S3 bucket. These .csv files must be converted into images and must be made available as soon as possible for the automatic generation of graphical reports.

The images become irrelevant after 1 month, but the .csv files must be kept to train machine learning (ML) models twice a year. The ML trainings and audits are planned weeks in advance.

Which combination of steps will meet these requirements MOST cost-effectively? (Choose two.)

一家製造公司擁有將 .csv 檔案上傳到 Amazon S3 儲存桶的機器感測器。這些 .csv 檔案必須轉換為影像，並且必須盡快提供以自動產生圖形報告。

影像在 1 個月後變得無關緊要，但必須保留 .csv 文件，以便每年訓練兩次機器學習 (ML) 模型。機器學習訓練和審核是提前幾週規劃的。

哪一種步驟組合能夠最經濟有效地滿足這些要求？（選兩個。）

【選項】

A. Launch an Amazon EC2 Spot Instance that downloads the .csv files every hour, generates the image files, and uploads the images to the S3 bucket.

B. Design an AWS Lambda function that converts the .csv files into images and stores the images in the S3 bucket. Invoke the Lambda function when a .csv file is uploaded.

C. Create S3 Lifecycle rules for .csv files and image files in the S3 bucket. Transition the .csv files from S3 Standard to S3 Glacier 1 day after they are uploaded. Expire the image files after 30 days.

D. Create S3 Lifecycle rules for .csv files and image files in the S3 bucket. Transition the .csv files from S3 Standard to S3 One Zone-Infrequent Access (S3 One Zone-IA) 1 day after they are uploaded. Expire the image files after 30 days.

E. Create S3 Lifecycle rules for .csv files and image files in the S3 bucket. Transition the .csv files from S3 Standard to S3 Standard-Infrequent Access (S3 Standard-IA) 1 day after they are uploaded. Keep the image files in Reduced Redundancy Storage (RRS).

【參考答案】BC

【題號】question 431

【題目】

A company has developed a new video game as a web application. The application is in a three-tier architecture in a VPC with Amazon RDS for MySQL in the database layer. Several players will compete concurrently online. The game's developers want to display a top-10 scoreboard in near-real time and offer the ability to stop and restore the game while preserving the current scores.

What should a solutions architect do to meet these requirements?

一家公司開發了一款新的視頻遊戲作為網頁應用程式。該應用程式採用 VPC 中的三層架構，資料庫層為 Amazon RDS for MySQL。多名選手將同時在線上比賽。遊戲的開發人員希望近乎即時地顯示前 10 名記分牌，並提供停止和恢復遊戲的能力，同時保留當前分數。

解決方案架構師應該怎麼做才能滿足這些要求？

【選項】

A.Set up an Amazon ElastiCache for Memcached cluster to cache the scores for the web application to display.

B.Set up an Amazon ElastiCache for Redis cluster to compute and cache the scores for the web application to display.

C.Place an Amazon CloudFront distribution in front of the web application to cache the scoreboard in a section of the application.

D.Create a read replica on Amazon RDS for MySQL to run queries to compute the scoreboard and serve the read traffic to the web application.

【參考答案】B

【題號】question 432

【題目】

An ecommerce company wants to use machine learning (ML) algorithms to build and train models. The company will use the models to visualize complex scenarios and to detect trends in customer data. The architecture team wants to integrate its ML models with a reporting platform to analyze the augmented data and use the data directly in its business intelligence dashboards.

Which solution will meet these requirements with the LEAST operational overhead?

一家電子商務公司希望使用機器學習 (ML) 演算法來建立和訓練模型。該公司將使用這些模型來視覺化複雜的場景並檢測客戶資料的趨勢。架構團隊希望將其 ML 模型與報告平台集成，以分析增強數據並直接在其商業智慧儀表板中使用數據。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

A.Use AWS Glue to create an ML transform to build and train models. Use Amazon OpenSearch Service to visualize the data.

B.Use Amazon SageMaker to build and train models. Use Amazon QuickSight to visualize the data.

C. Use a pre-built ML Amazon Machine Image (AMI) from the AWS Marketplace to build and train models. Use Amazon OpenSearch Service to visualize the data.

D. Use Amazon QuickSight to build and train models by using calculated fields. Use Amazon QuickSight to visualize the data.

【參考答案】B

【題號】question 433

【題目】

A company is running its production and nonproduction environment workloads in multiple AWS accounts. The accounts are in an organization in AWS Organizations. The company needs to design a solution that will prevent the modification of cost usage tags.

Which solution will meet these requirements?

一家公司在多個 AWS 帳戶中運行其生產和非生產環境工作負載。這些帳戶位於 AWS Organizations 中的組織中。該公司需要設計一種解決方案來防止成本使用標籤被修改。

哪種解決方案可以滿足這些要求？

【選項】

A. Create a custom AWS Config rule to prevent tag modification except by authorized principals.

B. Create a custom trail in AWS CloudTrail to prevent tag modification.

C. Create a service control policy (SCP) to prevent tag modification except by authorized principals.

D. Create custom Amazon CloudWatch logs to prevent tag modification.

【參考答案】C

【題號】question 434

【題目】

A company hosts its application in the AWS Cloud. The application runs on Amazon EC2 instances behind an Elastic Load Balancer in an Auto Scaling group and with an Amazon DynamoDB table. The company wants to ensure the application can be made available in another AWS Region with minimal downtime.

What should a solutions architect do to meet these requirements with the LEAST amount of downtime?

一家公司將其應用程式託管在 AWS 雲端。該應用程式在 Auto Scaling 群組中的 Elastic Load Balancer 後面的 Amazon EC2 執行個體上運行，並使用 Amazon DynamoDB 表。該公司希望確保該應用程式可以在另一個 AWS 區域中使用，並且停機時間最短。

解決方案架構師應該如何做才能以最少的停機時間來滿足這些要求？

【選項】

A. Create an Auto Scaling group and a load balancer in the disaster recovery Region.

Configure the DynamoDB table as a global table. Configure DNS failover to point to the new disaster recovery Region's load balancer.

B. Create an AWS CloudFormation template to create EC2 instances, load balancers, and DynamoDB tables to be launched when needed. Configure DNS failover to point to the new disaster recovery Region's load balancer.

C. Create an AWS CloudFormation template to create EC2 instances and a load balancer to be launched when needed. Configure the DynamoDB table as a global table. Configure DNS failover to point to the new disaster recovery Region's load balancer.

D. Create an Auto Scaling group and load balancer in the disaster recovery Region. Configure the DynamoDB table as a global table. Create an Amazon CloudWatch alarm to trigger an AWS Lambda function that updates Amazon Route 53 pointing to the disaster recovery load balancer.

【參考答案】A

【題號】question 435

【題目】

A company needs to migrate a MySQL database from its on-premises data center to AWS within 2 weeks. The database is 20 TB in size. The company wants to complete the migration with minimal downtime.

Which solution will migrate the database MOST cost-effectively?

一家公司需要在 2 週內將 MySQL 資料庫從其本地資料中心遷移到 AWS。該資料庫大小為 20 TB。該公司希望以最短的停機時間完成遷移。

哪一種解決方案能夠最經濟有效地遷移資料庫？

【選項】

A. Order an AWS Snowball Edge Storage Optimized device. Use AWS Database Migration Service (AWS DMS) with AWS Schema Conversion Tool (AWS SCT) to migrate the database with replication of ongoing changes. Send the Snowball Edge device to AWS to finish the migration and continue the ongoing replication.

B. Order an AWS Snowmobile vehicle. Use AWS Database Migration Service (AWS DMS) with AWS Schema Conversion Tool (AWS SCT) to migrate the database with ongoing changes. Send the Snowmobile vehicle back to AWS to finish the migration and continue the ongoing replication.

C. Order an AWS Snowball Edge Compute Optimized with GPU device. Use AWS Database Migration Service (AWS DMS) with AWS Schema Conversion Tool (AWS SCT) to migrate the database with ongoing changes. Send the Snowball device to AWS to finish the migration and continue the ongoing replication.

D. Order a 1 GB dedicated AWS Direct Connect connection to establish a connection with the data center. Use AWS Database Migration Service (AWS DMS) with AWS Schema Conversion Tool (AWS SCT) to migrate the database with replication of ongoing changes.

【參考答案】A

【題號】question 436

【題目】

A company moved its on-premises PostgreSQL database to an Amazon RDS for PostgreSQL DB instance. The company successfully launched a new product. The workload on the database has increased. The company wants to accommodate the larger workload without adding infrastructure.

Which solution will meet these requirements MOST cost-effectively?

一家公司將其本地 PostgreSQL 資料庫遷移到 Amazon RDS for PostgreSQL 資料庫執行個體。本公司成功推出新產品。資料庫的工作負載增加了。該公司希望在不增加基礎設施的情況下容納更大的工作負載。

哪一種解決方案能夠最具成本效益地滿足這些要求？

【選項】

- A. Buy reserved DB instances for the total workload. Make the Amazon RDS for PostgreSQL DB instance larger.
- B. Make the Amazon RDS for PostgreSQL DB instance a Multi-AZ DB instance.
- C. Buy reserved DB instances for the total workload. Add another Amazon RDS for PostgreSQL DB instance.
- D. Make the Amazon RDS for PostgreSQL DB instance an on-demand DB instance.

【參考答案】A

【題號】question 437

【題目】

A company operates an ecommerce website on Amazon EC2 instances behind an Application Load Balancer (ALB) in an Auto Scaling group. The site is experiencing performance issues related to a high request rate from illegitimate external systems with changing IP addresses. The security team is worried about potential DDoS attacks against

the website. The company must block the illegitimate incoming requests in a way that has a minimal impact on legitimate users.

What should a solutions architect recommend?

一家公司在 Auto Scaling 群組中的 Application Load Balancer (ALB) 後面的 Amazon EC2 執行個體上經營一個電子商務網站。該網站遇到與 IP 位址不斷變化的非法外部系統的高請求率相關的效能問題。安全團隊擔心該網站可能遭受 DDoS 攻擊。公司必須以對合法用戶影響最小的方式阻止非法傳入請求。

解決方案架構師該推薦什麼？

【選項】

A. Deploy Amazon Inspector and associate it with the ALB.

B. Deploy AWS WAF, associate it with the ALB, and configure a rate-limiting rule.

C. Deploy rules to the network ACLs associated with the ALB to block the incoming traffic.

D. Deploy Amazon GuardDuty and enable rate-limiting protection when configuring GuardDuty.

【參考答案】B

【題號】question 438

【題目】

A company wants to share accounting data with an external auditor. The data is stored in an Amazon RDS DB instance that resides in a private subnet. The auditor has its own AWS account and requires its own copy of the database.

What is the MOST secure way for the company to share the database with the auditor?

一家公司希望與外部審計師分享會計數據。資料儲存在位於私有子網路中的 Amazon RDS 資料庫執行個體中。審計員擁有自己的 AWS 帳戶，並需要自己的資料庫副本。

公司與審計師共享資料庫最安全的方式是什麼？

【選項】

A.Create a read replica of the database. Configure IAM standard database authentication to grant the auditor access.

B.Export the database contents to text files. Store the files in an Amazon S3 bucket. Create a new IAM user for the auditor. Grant the user access to the S3 bucket.

C.Copy a snapshot of the database to an Amazon S3 bucket. Create an IAM user. Share the user's keys with the auditor to grant access to the object in the S3 bucket.

D.Create an encrypted snapshot of the database. Share the snapshot with the auditor. Allow access to the AWS Key Management Service (AWS KMS) encryption key.

【參考答案】D

【題號】question 439

【題目】

A solutions architect configured a VPC that has a small range of IP addresses. The number of Amazon EC2 instances that are in the VPC is increasing, and there is an insufficient number of IP addresses for future workloads.

Which solution resolves this issue with the LEAST operational overhead?

解決方案架構師配置了一個具有小範圍 IP 位址的 VPC。VPC 中的 Amazon EC2 執行個體數量不斷增加，但 IP 位址數量不足以滿足未來的工作負載。

哪種解決方案能夠以最少的營運開銷解決此問題？

【選項】

A.Add an additional IPv4 CIDR block to increase the number of IP addresses and create additional subnets in the VPC. Create new resources in the new subnets by using the new CIDR.

B.Create a second VPC with additional subnets. Use a peering connection to connect the second VPC with the first VPC. Update the routes and create new resources in the subnets of the second VPC.

C. Use AWS Transit Gateway to add a transit gateway and connect a second VPC with the first VPC. Update the routes of the transit gateway and VPCs. Create new resources in the subnets of the second VPC.

D. Create a second VPC. Create a Site-to-Site VPN connection between the first VPC and the second VPC by using a VPN-hosted solution on Amazon EC2 and a virtual private gateway. Update the route between VPCs to the traffic through the VPN. Create new resources in the subnets of the second VPC.

【參考答案】A

【題號】question 440

【題目】

A company used an Amazon RDS for MySQL DB instance during application testing. Before terminating the DB instance at the end of the test cycle, a solutions architect created two backups. The solutions architect created the first backup by using the mysqldump utility to create a database dump. The solutions architect created the second backup by enabling the final DB snapshot option on RDS termination.

The company is now planning for a new test cycle and wants to create a new DB instance from the most recent backup. The company has chosen a MySQL-compatible edition of Amazon Aurora to host the DB instance.

Which solutions will create the new DB instance? (Choose two.)

一家公司在應用程式測試期間使用 Amazon RDS for MySQL 資料庫執行個體。在測試週期結束時終止資料庫執行個體之前，解決方案架構師建立了兩個備份。解決方案架構師透過使用 mysqldump 實用程式建立資料庫轉儲來建立第一個備份。解決方案架構師透過在 RDS 終止時啟用最終資料庫快照選項來建立第二個備份。

該公司現在正在計劃一個新的測試週期，並希望根據最新的備份建立一個新的資料庫實例。該公司選擇了與 MySQL 相容的 Amazon Aurora 版本來託管資料庫執行個體。

哪些解決方案將建立新的資料庫實例？（選兩個。）

【選項】

A. Import the RDS snapshot directly into Aurora.

- B.Upload the RDS snapshot to Amazon S3. Then import the RDS snapshot into Aurora.
- C.Upload the database dump to Amazon S3. Then import the database dump into Aurora.
- D.Use AWS Database Migration Service (AWS DMS) to import the RDS snapshot into Aurora.
- E.Upload the database dump to Amazon S3. Then use AWS Database Migration Service (AWS DMS) to import the database dump into Aurora.

【參考答案】AC

【題號】question 441

【題目】

A company hosts a multi-tier web application on Amazon Linux Amazon EC2 instances behind an Application Load Balancer. The instances run in an Auto Scaling group across multiple Availability Zones. The company observes that the Auto Scaling group launches more On-Demand Instances when the application's end users access high volumes of static web content. The company wants to optimize cost.

What should a solutions architect do to redesign the application MOST cost-effectively?

一家公司在應用程式負載平衡器後面的 Amazon Linux Amazon EC2 執行個體上託管多層 Web 應用程式。執行個體在跨多個可用區的 Auto Scaling 群組中執行。該公司觀察到，當應用程式的最終使用者存取大量靜態 Web 內容時，Auto Scaling 小組會啟動更多的按需實例。該公司希望優化成本。

解決方案架構師應該如何以最經濟有效的方式重新設計應用程式？

【選項】

- A.Update the Auto Scaling group to use Reserved Instances instead of On-Demand Instances.
- B.Update the Auto Scaling group to scale by launching Spot Instances instead of On-Demand Instances.
- C.Create an Amazon CloudFront distribution to host the static web contents from an Amazon S3 bucket.

D.Create an AWS Lambda function behind an Amazon API Gateway API to host the static website contents.

【參考答案】C

【題號】question 442

【題目】

A company stores several petabytes of data across multiple AWS accounts. The company uses AWS Lake Formation to manage its data lake. The company's data science team wants to securely share selective data from its accounts with the company's engineering team for analytical purposes.

Which solution will meet these requirements with the LEAST operational overhead?

一家公司在多個 AWS 帳戶中儲存數 PB 的資料。該公司使用 AWS Lake Formation 來管理其資料湖。該公司的數據科學團隊希望與該公司的工程團隊安全地共享其帳戶中的選擇性數據以進行分析。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

A.Copy the required data to a common account. Create an IAM access role in that account. Grant access by specifying a permission policy that includes users from the engineering team accounts as trusted entities.

B.Use the Lake Formation permissions Grant command in each account where the data is stored to allow the required engineering team users to access the data.

C.Use AWS Data Exchange to privately publish the required data to the required engineering team accounts.

D.Use Lake Formation tag-based access control to authorize and grant cross-account permissions for the required data to the engineering team accounts.

【參考答案】D

【題號】question 443

【題目】

A company wants to host a scalable web application on AWS. The application will be accessed by users from different geographic regions of the world. Application users will be able to download and upload unique data up to gigabytes in size. The development team wants a cost-effective solution to minimize upload and download latency and maximize performance.

What should a solutions architect do to accomplish this?

一家公司希望在 AWS 上託管可擴展的 Web 應用程式。來自世界不同地理區域的用戶將存取該應用程式。應用程式用戶將能夠下載和上傳高達千兆位元組的獨特資料。開發團隊需要一個經濟高效的解決方案來最大限度地減少上傳和下載延遲並最大限度地提高效能。

解決方案架構師應該做什麼來實現這個目標？

【選項】

- A. Use Amazon S3 with Transfer Acceleration to host the application.
- B. Use Amazon S3 with CacheControl headers to host the application.
- C. Use Amazon EC2 with Auto Scaling and Amazon CloudFront to host the application.
- D. Use Amazon EC2 with Auto Scaling and Amazon ElastiCache to host the application.

【參考答案】A

【題號】question 444

【題目】

A company has hired a solutions architect to design a reliable architecture for its application. The application consists of one Amazon RDS DB instance and two manually

provisioned Amazon EC2 instances that run web servers. The EC2 instances are located in a single Availability Zone.

An employee recently deleted the DB instance, and the application was unavailable for 24 hours as a result. The company is concerned with the overall reliability of its environment.

What should the solutions architect do to maximize reliability of the application's infrastructure?

一家公司聘請了一位解決方案架構師為其應用程式設計可靠的架構。該應用程式由一個 Amazon RDS 資料庫執行個體和兩個手動配置的運行 Web 伺服器的 Amazon EC2 執行個體組成。EC2 執行個體位於單一可用區。

一名員工最近刪除了資料庫實例，導致應用程式 24 小時不可用。該公司關心其環境的整體可靠性。

解決方案架構師應該怎麼做才能最大限度地提高應用程式基礎架構的可靠性？

【選項】

A.Delete one EC2 instance and enable termination protection on the other EC2 instance. Update the DB instance to be Multi-AZ, and enable deletion protection.

B.Update the DB instance to be Multi-AZ, and enable deletion protection. Place the EC2 instances behind an Application Load Balancer, and run them in an EC2 Auto Scaling group across multiple Availability Zones.

C.Create an additional DB instance along with an Amazon API Gateway and an AWS Lambda function. Configure the application to invoke the Lambda function through API Gateway. Have the Lambda function write the data to the two DB instances.

D.Place the EC2 instances in an EC2 Auto Scaling group that has multiple subnets located in multiple Availability Zones. Use Spot Instances instead of On-Demand Instances. Set up Amazon CloudWatch alarms to monitor the health of the instances Update the DB instance to be Multi-AZ, and enable deletion protection.

【參考答案】B

【題號】question 445

【題目】

A company is storing 700 terabytes of data on a large network-attached storage (NAS) system in its corporate data center. The company has a hybrid environment with a 10 Gbps AWS Direct Connect connection.

After an audit from a regulator, the company has 90 days to move the data to the cloud. The company needs to move the data efficiently and without disruption. The company still needs to be able to access and update the data during the transfer window.

Which solution will meet these requirements?

一家公司正在其公司資料中心的大型網路附加儲存 (NAS) 系統上儲存 700 TB 的資料。該公司擁有一個具有 10 Gbps AWS Direct Connect 連線的混合環境。

經過監管機構的審核後，該公司有 90 天的時間將資料轉移到雲端。公司需要有效率且不間斷地移動數據。該公司仍然需要能夠在轉移視窗期間存取和更新資料。

哪種解決方案可以滿足這些要求？

【選項】

A.Create an AWS DataSync agent in the corporate data center. Create a data transfer task. Start the transfer to an Amazon S3 bucket.

B.Back up the data to AWS Snowball Edge Storage Optimized devices. Ship the devices to an AWS data center. Mount a target Amazon S3 bucket on the on-premises file system.

C.Use rsync to copy the data directly from local storage to a designated Amazon S3 bucket over the Direct Connect connection.

D.Back up the data on tapes. Ship the tapes to an AWS data center. Mount a target Amazon S3 bucket on the on-premises file system.

【參考答案】A

【題號】question 446

【題目】

A company stores data in PDF format in an Amazon S3 bucket. The company must follow a legal requirement to retain all new and existing data in Amazon S3 for 7 years.

Which solution will meet these requirements with the LEAST operational overhead?

一家公司將 PDF 格式的資料儲存在 Amazon S3 儲存桶中。公司必須遵守法律要求，將所有新資料和現有資料在 Amazon S3 中保留 7 年。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

A. Turn on the S3 Versioning feature for the S3 bucket. Configure S3 Lifecycle to delete the data after 7 years. Configure multi-factor authentication (MFA) delete for all S3 objects.

B. Turn on S3 Object Lock with governance retention mode for the S3 bucket. Set the retention period to expire after 7 years. Recopy all existing objects to bring the existing data into compliance.

C. Turn on S3 Object Lock with compliance retention mode for the S3 bucket. Set the retention period to expire after 7 years. Recopy all existing objects to bring the existing data into compliance.

D. Turn on S3 Object Lock with compliance retention mode for the S3 bucket. Set the retention period to expire after 7 years. Use S3 Batch Operations to bring the existing data into compliance.

【參考答案】D

【題號】question 447

【題目】

A company has a stateless web application that runs on AWS Lambda functions that are invoked by Amazon API Gateway. The company wants to deploy the application across multiple AWS Regions to provide Regional failover capabilities.

What should a solutions architect do to route traffic to multiple Regions?

一家公司有一個無狀態 Web 應用程式，該應用程式在 Amazon API Gateway 呼叫的 AWS Lambda 函數上運行。該公司希望跨多個 AWS 區域部署應用程式以提供區域故障轉移功能。

解決方案架構師應該如何將流量路由到多個區域？

【選項】

A.Create Amazon Route 53 health checks for each Region. Use an active-active failover configuration.

B.Create an Amazon CloudFront distribution with an origin for each Region. Use CloudFront health checks to route traffic.

C.Create a transit gateway. Attach the transit gateway to the API Gateway endpoint in each Region. Configure the transit gateway to route requests.

D.Create an Application Load Balancer in the primary Region. Set the target group to point to the API Gateway endpoint hostnames in each Region.

【參考答案】A

【題號】question 448

【題目】

A company has two VPCs named Management and Production. The Management VPC uses VPNs through a customer gateway to connect to a single device in the data center. The Production VPC uses a virtual private gateway with two attached AWS Direct Connect connections. The Management and Production VPCs both use a single VPC peering connection to allow communication between the applications.

What should a solutions architect do to mitigate any single point of failure in this architecture?

某公司有兩個 VPC，分別為管理和生產。管理 VPC 透過客戶閘道使用 VPN 連線到資料中心中的單一裝置。生產 VPC 使用具有兩個附加 AWS Direct Connect 連線的虛擬私有閘道。管理 VPC 和生產 VPC 均使用單一 VPC 對等連接來允許應用程式之間進行通訊。

解決方案架構師應該採取什麼措施來減輕該架構中的任何單點故障？

【選項】

- A.Add a set of VPNs between the Management and Production VPCs.
- B.Add a second virtual private gateway and attach it to the Management VPC.
- C.Add a second set of VPNs to the Management VPC from a second customer gateway device.
- D.Add a second VPC peering connection between the Management VPC and the Production VPC.

【參考答案】C

【題號】question 449

【題目】

A company runs its application on an Oracle database. The company plans to quickly migrate to AWS because of limited resources for the database, backup administration, and data center maintenance. The application uses third-party database features that require privileged access.

Which solution will help the company migrate the database to AWS MOST cost-effectively?

一家公司在 Oracle 資料庫上運行其應用程式。由於資料庫、備份管理和資料中心維護的資源有限，該公司計劃快速遷移到 AWS。該應用程式使用需要特權存取的第三方資料庫功能。

哪種解決方案將幫助公司最經濟有效地將資料庫遷移到 AWS？

【選項】

- A.Migrate the database to Amazon RDS for Oracle. Replace third-party features with cloud services.
- B.Migrate the database to Amazon RDS Custom for Oracle. Customize the database settings to support third-party features.
- C.Migrate the database to an Amazon EC2 Amazon Machine Image (AMI) for Oracle. Customize the database settings to support third-party features.

D.Migrate the database to Amazon RDS for PostgreSQL by rewriting the application code to remove dependency on Oracle APEX.

【參考答案】B

【題號】question 450

【題目】

A company has a three-tier web application that is in a single server. The company wants to migrate the application to the AWS Cloud. The company also wants the application to align with the AWS Well-Architected Framework and to be consistent with AWS recommended best practices for security, scalability, and resiliency.

Which combination of solutions will meet these requirements? (Choose three.)

一家公司有一個位於單一伺服器中的三層 Web 應用程式。該公司希望將應用程式遷移到 AWS 雲端。該公司還希望該應用程式與 AWS Well-Architected Framework 保持一致，並與 AWS 推薦的安全性、可擴展性和彈性最佳實踐保持一致。

哪種解決方案組合可以滿足這些要求？（選擇三項。）

【選項】

A.Create a VPC across two Availability Zones with the application's existing architecture. Host the application with existing architecture on an Amazon EC2 instance in a private subnet in each Availability Zone with EC2 Auto Scaling groups. Secure the EC2 instance with security groups and network access control lists (network ACLs).

B.Set up security groups and network access control lists (network ACLs) to control access to the database layer. Set up a single Amazon RDS database in a private subnet.

C.Create a VPC across two Availability Zones. Refactor the application to host the web tier, application tier, and database tier. Host each tier on its own private subnet with Auto Scaling groups for the web tier and application tier.

D.Use a single Amazon RDS database. Allow database access only from the application tier security group.

E.Use Elastic Load Balancers in front of the web tier. Control access by using security groups containing references to each layer's security groups.

F.Use an Amazon RDS database Multi-AZ cluster deployment in private subnets. Allow database access only from application tier security groups.

【參考答案】CEF

【題號】question 451

【題目】

A company is migrating its applications and databases to the AWS Cloud. The company will use Amazon Elastic Container Service (Amazon ECS), AWS Direct Connect, and Amazon RDS.

Which activities will be managed by the company's operational team? (Choose three.)

一家公司正在將其應用程式和資料庫遷移到 AWS 雲端。該公司將使用 Amazon Elastic Container Service (Amazon ECS)、AWS Direct Connect 和 Amazon RDS。

哪些活動將由公司的營運團隊管理？（選擇三項。）

【選項】

A.Management of the Amazon RDS infrastructure layer, operating system, and platforms

B.Creation of an Amazon RDS DB instance and configuring the scheduled maintenance window

C.Configuration of additional software components on Amazon ECS for monitoring, patch management, log management, and host intrusion detection

D.Installation of patches for all minor and major database versions for Amazon RDS

E.Ensure the physical security of the Amazon RDS infrastructure in the data center

F.Encryption of the data that moves in transit through Direct Connect

【參考答案】BCF

【題號】question 452

【題目】

A company runs a Java-based job on an Amazon EC2 instance. The job runs every hour and takes 10 seconds to run. The job runs on a scheduled interval and consumes 1 GB of memory. The CPU utilization of the instance is low except for short surges during which the job uses the maximum CPU available. The company wants to optimize the costs to run the job.

Which solution will meet these requirements?

一家公司在 Amazon EC2 執行個體上執行基於 Java 的作業。此作業每小時運行一次，運行時間為 10 秒。此作業會依照排程的時間間隔運作並消耗 1 GB 記憶體。除了作業使用最大可用 CPU 的短暫激增之外，執行個體的 CPU 使用率較低。該公司希望優化這項工作的成本。

哪種解決方案可以滿足這些要求？

【選項】

A. Use AWS App2Container (A2C) to containerize the job. Run the job as an Amazon Elastic Container Service (Amazon ECS) task on AWS Fargate with 0.5 virtual CPU (vCPU) and 1 GB of memory.

B. Copy the code into an AWS Lambda function that has 1 GB of memory. Create an Amazon EventBridge scheduled rule to run the code each hour.

C. Use AWS App2Container (A2C) to containerize the job. Install the container in the existing Amazon Machine Image (AMI). Ensure that the schedule stops the container when the task finishes.

D. Configure the existing schedule to stop the EC2 instance at the completion of the job and restart the EC2 instance when the next job starts.

【參考答案】B

【題號】question 453

【題目】

A company wants to implement a backup strategy for Amazon EC2 data and multiple Amazon S3 buckets. Because of regulatory requirements, the company must retain backup files for a specific time period. The company must not alter the files for the duration of the retention period.

Which solution will meet these requirements?

一家公司想要為 Amazon EC2 資料和多個 Amazon S3 儲存桶實施備份策略。由於法規要求，公司必須將備份文件保留特定時間段。在保留期內，公司不得更改文件。

哪種解決方案可以滿足這些要求？

【選項】

A. Use AWS Backup to create a backup vault that has a vault lock in governance mode. Create the required backup plan.

B. Use Amazon Data Lifecycle Manager to create the required automated snapshot policy.

C. Use Amazon S3 File Gateway to create the backup. Configure the appropriate S3 Lifecycle management.

D. Use AWS Backup to create a backup vault that has a vault lock in compliance mode. Create the required backup plan.

【參考答案】D

【題號】question 454

【題目】

A company has resources across multiple AWS Regions and accounts. A newly hired solutions architect discovers a previous employee did not provide details about the resources inventory. The solutions architect needs to build and map the relationship details of the various workloads across all accounts.

Which solution will meet these requirements in the MOST operationally efficient way?

一家公司擁有跨多個 AWS 區域和帳戶的資源。新聘用的解決方案架構師發現以前的員工沒有提供有關資源庫存的詳細資訊。解決方案架構師需要建立和映射所有帳戶中各種工作負載的關係詳細資訊。

哪一種解決方案能夠以最高效的方式滿足這些要求？

【選項】

A. Use AWS Systems Manager Inventory to generate a map view from the detailed view report.

B. Use AWS Step Functions to collect workload details. Build architecture diagrams of the workloads manually.

C. Use Workload Discovery on AWS to generate architecture diagrams of the workloads.

D. Use AWS X-Ray to view the workload details. Build architecture diagrams with relationships.

【參考答案】C

【題號】question 455

【題目】

A company uses AWS Organizations. The company wants to operate some of its AWS accounts with different budgets. The company wants to receive alerts and automatically prevent provisioning of additional resources on AWS accounts when the allocated budget threshold is met during a specific period.

Which combination of solutions will meet these requirements? (Choose three.)

一家公司使用 AWS Organizations。該公司希望以不同的預算來營運其一些 AWS 帳戶。該公司希望在特定時間段內達到分配的預算閾值時接收警報並自動阻止在 AWS 帳戶上配置額外資源。

哪種解決方案組合可以滿足這些要求？（選擇三項。）

【選項】

A. Use AWS Budgets to create a budget. Set the budget amount under the Cost and Usage Reports section of the required AWS accounts.

B. Use AWS Budgets to create a budget. Set the budget amount under the Billing dashboards of the required AWS accounts.

C. Create an IAM user for AWS Budgets to run budget actions with the required permissions.

D. Create an IAM role for AWS Budgets to run budget actions with the required permissions.

E. Add an alert to notify the company when each account meets its budget threshold. Add a budget action that selects the IAM identity created with the appropriate config rule to prevent provisioning of additional resources.

F. Add an alert to notify the company when each account meets its budget threshold. Add a budget action that selects the IAM identity created with the appropriate service control policy (SCP) to prevent provisioning of additional resources.

【參考答案】BDF

【題號】question 456

【題目】

A company runs applications on Amazon EC2 instances in one AWS Region. The company wants to back up the EC2 instances to a second Region. The company also wants to provision EC2 resources in the second Region and manage the EC2 instances centrally from one AWS account.

Which solution will meet these requirements MOST cost-effectively?

一家公司在一個 AWS 區域的 Amazon EC2 執行個體上執行應用程式。該公司希望將 EC2 執行個體備份到第二個區域。該公司還希望在第二個區域配置 EC2 資源，並從一個 AWS 帳戶集中管理 EC2 執行個體。

哪一種解決方案能夠最具成本效益地滿足這些要求？

【選項】

A.Create a disaster recovery (DR) plan that has a similar number of EC2 instances in the second Region. Configure data replication.

B.Create point-in-time Amazon Elastic Block Store (Amazon EBS) snapshots of the EC2 instances. Copy the snapshots to the second Region periodically.

C.Create a backup plan by using AWS Backup. Configure cross-Region backup to the second Region for the EC2 instances.

D.Deploy a similar number of EC2 instances in the second Region. Use AWS DataSync to transfer the data from the source Region to the second Region.

【參考答案】C

【題號】question 457

【題目】

A company that uses AWS is building an application to transfer data to a product manufacturer. The company has its own identity provider (IdP). The company wants the IdP to authenticate application users while the users use the application to transfer data. The company must use Applicability Statement 2 (AS2) protocol.

Which solution will meet these requirements?

一家使用 AWS 的公司正在建立一個應用程式來將資料傳輸給產品製造商。該公司擁有自己的身分提供者 (IdP)。該公司希望 IdP 在用戶使用應用程式傳輸資料時對應用程式使用者進行身份驗證。公司必須使用適用性聲明 2 (AS2) 協議。

哪種解決方案可以滿足這些要求？

【選項】

A.Use AWS DataSync to transfer the data. Create an AWS Lambda function for IdP authentication.

B.Use Amazon AppFlow flows to transfer the data. Create an Amazon Elastic Container Service (Amazon ECS) task for IdP authentication.

C.Use AWS Transfer Family to transfer the data. Create an AWS Lambda function for IdP authentication.

D. Use AWS Storage Gateway to transfer the data. Create an Amazon Cognito identity pool for IdP authentication.

【參考答案】C

【題號】question 458

【題目】

A solutions architect is designing a REST API in Amazon API Gateway for a cash payback service. The application requires 1 GB of memory and 2 GB of storage for its computation resources. The application will require that the data is in a relational format.

Which additional combination of AWS services will meet these requirements with the LEAST administrative effort? (Choose two.)

解決方案架構師正在 Amazon API Gateway 中設計 REST API 以實現現金回報服務。該應用程式需要 1 GB 記憶體和 2 GB 儲存空間用於其計算資源。應用程式將要求資料採用關係格式。

哪一種額外的 AWS 服務組合能夠以最少的管理工作滿足這些要求？（選兩個。）

【選項】

- A. Amazon EC2
- B. AWS Lambda
- C. Amazon RDS
- D. Amazon DynamoDB
- E. Amazon Elastic Kubernetes Services (Amazon EKS)

【參考答案】BC

【題號】question 459

【題目】

A company uses AWS Organizations to run workloads within multiple AWS accounts. A tagging policy adds department tags to AWS resources when the company creates tags.

An accounting team needs to determine spending on Amazon EC2 consumption. The accounting team must determine which departments are responsible for the costs regardless of AWS account. The accounting team has access to AWS Cost Explorer for all AWS accounts within the organization and needs to access all reports from Cost Explorer.

Which solution meets these requirements in the MOST operationally efficient way?

一家公司使用 AWS Organizations 在多個 AWS 帳戶中執行工作負載。當公司建立標籤時，標記策略會將部門標籤新增至 AWS 資源。

會計團隊需要確定 Amazon EC2 消耗的支出。會計團隊必須確定哪些部門負責成本，無論 AWS 帳戶為何。會計團隊可以存取組織內所有 AWS 帳戶的 AWS Cost Explorer，並且需要存取 Cost Explorer 中的所有報告。

哪種解決方案以最高效的方式滿足這些要求？

【選項】

A.From the Organizations management account billing console, activate a user-defined cost allocation tag named department. Create one cost report in Cost Explorer grouping by tag name, and filter by EC2.

B.From the Organizations management account billing console, activate an AWS-defined cost allocation tag named department. Create one cost report in Cost Explorer grouping by tag name, and filter by EC2.

C.From the Organizations member account billing console, activate a user-defined cost allocation tag named department. Create one cost report in Cost Explorer grouping by the tag name, and filter by EC2.

D.From the Organizations member account billing console, activate an AWS-defined cost allocation tag named department. Create one cost report in Cost Explorer grouping by tag name, and filter by EC2.

【參考答案】A

【題號】question 460

【題目】

A company wants to securely exchange data between its software as a service (SaaS) application Salesforce account and Amazon S3. The company must encrypt the data at rest by using AWS Key Management Service (AWS KMS) customer managed keys (CMKs). The company must also encrypt the data in transit. The company has enabled API access for the Salesforce account.

一家公司希望在其軟體即服務 (SaaS) 應用程式 Salesforce 帳戶和 Amazon S3 之間安全地交換資料。公司必須使用 AWS Key Management Service (AWS KMS) 客戶管理金鑰 (CMK) 加密靜態資料。公司還必須對傳輸中的資料進行加密。該公司已為 Salesforce 帳戶啟用 API 存取。

【選項】

- A.Create AWS Lambda functions to transfer the data securely from Salesforce to Amazon S3.
- B.Create an AWS Step Functions workflow. Define the task to transfer the data securely from Salesforce to Amazon S3.
- C.Create Amazon AppFlow flows to transfer the data securely from Salesforce to Amazon S3.
- D.Create a custom connector for Salesforce to transfer the data securely from Salesforce to Amazon S3.

【參考答案】C

【題號】question 461

【題目】

A company is developing a mobile gaming app in a single AWS Region. The app runs on multiple Amazon EC2 instances in an Auto Scaling group. The company stores the app data in Amazon DynamoDB. The app communicates by using TCP traffic and UDP traffic

between the users and the servers. The application will be used globally. The company wants to ensure the lowest possible latency for all users.

Which solution will meet these requirements?

一家公司正在單一 AWS 區域中開發行動遊戲應用程式。該應用程式在 Auto Scaling 群組中的多個 Amazon EC2 執行個體上執行。該公司將應用程式資料儲存在 Amazon DynamoDB 中。該應用程式透過使用用戶和伺服器之間的 TCP 流量和 UDP 流量進行通訊。該應用程式將在全球範圍內使用。該公司希望確保所有用戶的延遲盡可能最低。

哪種解決方案可以滿足這些要求？

【選項】

A. Use AWS Global Accelerator to create an accelerator. Create an Application Load Balancer (ALB) behind an accelerator endpoint that uses Global Accelerator integration and listening on the TCP and UDP ports. Update the Auto Scaling group to register instances on the ALB.

B. Use AWS Global Accelerator to create an accelerator. Create a Network Load Balancer (NLB) behind an accelerator endpoint that uses Global Accelerator integration and listening on the TCP and UDP ports. Update the Auto Scaling group to register instances on the NLB.

C. Create an Amazon CloudFront content delivery network (CDN) endpoint. Create a Network Load Balancer (NLB) behind the endpoint and listening on the TCP and UDP ports. Update the Auto Scaling group to register instances on the NLB. Update CloudFront to use the NLB as the origin.

D. Create an Amazon CloudFront content delivery network (CDN) endpoint. Create an Application Load Balancer (ALB) behind the endpoint and listening on the TCP and UDP ports. Update the Auto Scaling group to register instances on the ALB. Update CloudFront to use the ALB as the origin.

【參考答案】B

【題號】question 462

【題目】

A company has an application that processes customer orders. The company hosts the application on an Amazon EC2 instance that saves the orders to an Amazon Aurora database. Occasionally when traffic is high the workload does not process orders fast enough.

What should a solutions architect do to write the orders reliably to the database as quickly as possible?

一家公司有一個處理客戶訂單的應用程式。該公司將該應用程式託管在 Amazon EC2 執行個體上，該執行個體將訂單儲存到 Amazon Aurora 資料庫中。有時，當流量很高時，工作負載處理訂單的速度不夠快。

解決方案架構師該如何做才能盡快將訂單可靠地寫入資料庫？

【選項】

A.Increase the instance size of the EC2 instance when traffic is high. Write orders to Amazon Simple Notification Service (Amazon SNS). Subscribe the database endpoint to the SNS topic.

B.Write orders to an Amazon Simple Queue Service (Amazon SQS) queue. Use EC2 instances in an Auto Scaling group behind an Application Load Balancer to read from the SQS queue and process orders into the database.

C.Write orders to Amazon Simple Notification Service (Amazon SNS). Subscribe the database endpoint to the SNS topic. Use EC2 instances in an Auto Scaling group behind an Application Load Balancer to read from the SNS topic.

D.Write orders to an Amazon Simple Queue Service (Amazon SQS) queue when the EC2 instance reaches CPU threshold limits. Use scheduled scaling of EC2 instances in an Auto Scaling group behind an Application Load Balancer to read from the SQS queue and process orders into the database.

【參考答案】B

【題號】question 463

【題目】

An IoT company is releasing a mattress that has sensors to collect data about a user's sleep. The sensors will send data to an Amazon S3 bucket. The sensors collect approximately 2 MB of data every night for each mattress. The company must process and summarize the data for each mattress. The results need to be available as soon as possible. Data processing will require 1 GB of memory and will finish within 30 seconds.

Which solution will meet these requirements MOST cost-effectively?

一家物聯網公司正在發布一款帶有感測器的床墊，可以收集有關用戶睡眠的數據。感測器將資料傳送到 Amazon S3 儲存桶。感測器每晚為每個床墊收集約 2 MB 的數據。公司必須處理和匯總每個床墊的數據。結果需要盡快提供。資料處理需要 1 GB 內存，並將在 30 秒內完成。

哪一種解決方案能夠最具成本效益地滿足這些要求？

【選項】

- A. Use AWS Glue with a Scala job
- B. Use Amazon EMR with an Apache Spark script
- C. Use AWS Lambda with a Python script
- D. Use AWS Glue with a PySpark job

【參考答案】C

【題號】question 464

【題目】

A company hosts an online shopping application that stores all orders in an Amazon RDS for PostgreSQL Single-AZ DB instance. Management wants to eliminate single points of failure and has asked a solutions architect to recommend an approach to minimize database downtime without requiring any changes to the application code.

Which solution meets these requirements?

一家公司託管一個線上購物應用程式，該應用程式將所有訂單儲存在 Amazon RDS for PostgreSQL 單可用區資料庫實例中。管理層希望消除單點故障，並要求解決方案架構師推薦一種方法來最大限度地減少資料庫停機時間，而無需對應用程式程式碼進行任何更改。

哪種解決方案滿足這些要求？

【選項】

A.Convert the existing database instance to a Multi-AZ deployment by modifying the database instance and specifying the Multi-AZ option.

B.Create a new RDS Multi-AZ deployment. Take a snapshot of the current RDS instance and restore the new Multi-AZ deployment with the snapshot.

C.Create a read-only replica of the PostgreSQL database in another Availability Zone. Use Amazon Route 53 weighted record sets to distribute requests across the databases.

D.Place the RDS for PostgreSQL database in an Amazon EC2 Auto Scaling group with a minimum group size of two. Use Amazon Route 53 weighted record sets to distribute requests across instances.

【參考答案】A

【題號】question 465

【題目】

A company is developing an application to support customer demands. The company wants to deploy the application on multiple Amazon EC2 Nitro-based instances within the same Availability Zone. The company also wants to give the application the ability to write to multiple block storage volumes in multiple EC2 Nitro-based instances simultaneously to achieve higher application availability.

Which solution will meet these requirements?

一家公司正在開發一款應用程式來支援客戶需求。該公司希望將該應用程式部署在同一可用區內的多個基於 Amazon EC2 Nitro 的執行個體上。該公司還希望使應用程式能夠同時寫入多個基於 EC2 Nitro 的執行個體中的多個區塊儲存卷，以實現更高的應用程式可用性。

哪種解決方案可以滿足這些要求？

【選項】

- A. Use General Purpose SSD (gp3) EBS volumes with Amazon Elastic Block Store (Amazon EBS) Multi-Attach
- B. Use Throughput Optimized HDD (st1) EBS volumes with Amazon Elastic Block Store (Amazon EBS) Multi-Attach
- C. Use Provisioned IOPS SSD (io2) EBS volumes with Amazon Elastic Block Store (Amazon EBS) Multi-Attach
- D. Use General Purpose SSD (gp2) EBS volumes with Amazon Elastic Block Store (Amazon EBS) Multi-Attach

【參考答案】C

【題號】question 466

【題目】

A company designed a stateless two-tier application that uses Amazon EC2 in a single Availability Zone and an Amazon RDS Multi-AZ DB instance. New company management wants to ensure the application is highly available.

What should a solutions architect do to meet this requirement?

一家公司設計了一個無狀態兩層應用程式，該應用程式在單一可用區中使用 Amazon EC2 和 Amazon RDS 多可用區資料庫執行個體。新公司管理層希望確保應用程式具有高可用性。

解決方案架構師應該怎麼做才能滿足這項要求？

【選項】

- A. Configure the application to use Multi-AZ EC2 Auto Scaling and create an Application Load Balancer
- B. Configure the application to take snapshots of the EC2 instances and send them to a different AWS Region
- C. Configure the application to use Amazon Route 53 latency-based routing to feed requests to the application

D.Configure Amazon Route 53 rules to handle incoming requests and create a Multi-AZ Application Load Balancer

【參考答案】A

【題號】question 467

【題目】

A company uses AWS Organizations. A member account has purchased a Compute Savings Plan. Because of changes in the workloads inside the member account, the account no longer receives the full benefit of the Compute Savings Plan commitment. The company uses less than 50% of its purchased compute power.

一家公司使用 AWS Organizations。會員帳戶已購買計算儲蓄計劃。由於成員帳戶內工作負載的變化，該帳戶不再獲得計算儲蓄計劃承諾的全部好處。該公司使用的運算能力不到其購買的 50%。

【選項】

- A.Turn on discount sharing from the Billing Preferences section of the account console in the member account that purchased the Compute Savings Plan.
- B.Turn on discount sharing from the Billing Preferences section of the account console in the company's Organizations management account.
- C.Migrate additional compute workloads from another AWS account to the account that has the Compute Savings Plan.
- D.Sell the excess Savings Plan commitment in the Reserved Instance Marketplace.

【參考答案】B

【題號】question 468

【題目】

A company is developing a microservices application that will provide a search catalog for customers. The company must use REST APIs to present the frontend of the application to users. The REST APIs must access the backend services that the company hosts in containers in private VPC subnets.

Which solution will meet these requirements?

一家公司正在開發一款微服務應用程式，該應用程式將為客戶提供搜尋目錄。公司必須使用 REST API 向使用者呈現應用程式的前端。REST API 必須存取公司在私有 VPC 子網路中的容器中託管的後端服務。

哪種解決方案可以滿足這些要求？

【選項】

A.Design a WebSocket API by using Amazon API Gateway. Host the application in Amazon Elastic Container Service (Amazon ECS) in a private subnet. Create a private VPC link for API Gateway to access Amazon ECS.

B.Design a REST API by using Amazon API Gateway. Host the application in Amazon Elastic Container Service (Amazon ECS) in a private subnet. Create a private VPC link for API Gateway to access Amazon ECS.

C.Design a WebSocket API by using Amazon API Gateway. Host the application in Amazon Elastic Container Service (Amazon ECS) in a private subnet. Create a security group for API Gateway to access Amazon ECS.

D.Design a REST API by using Amazon API Gateway. Host the application in Amazon Elastic Container Service (Amazon ECS) in a private subnet. Create a security group for API Gateway to access Amazon ECS.

【參考答案】B

【題號】question 469

【題目】

A company stores raw collected data in an Amazon S3 bucket. The data is used for several types of analytics on behalf of the company's customers. The type of analytics requested determines the access pattern on the S3 objects.

The company cannot predict or control the access pattern. The company wants to reduce its S3 costs.

Which solution will meet these requirements?

一家公司將收集的原始資料儲存在 Amazon S3 儲存桶中。這些數據代表公司客戶用於多種類型的分析。請求的分析類型決定了 S3 物件的存取模式。

公司無法預測或控制存取模式。該公司希望降低 S3 成本。

哪種解決方案可以滿足這些要求？

【選項】

A. Use S3 replication to transition infrequently accessed objects to S3 Standard-Infrequent Access (S3 Standard-IA)

B. Use S3 Lifecycle rules to transition objects from S3 Standard to Standard-Infrequent Access (S3 Standard-IA)

C. Use S3 Lifecycle rules to transition objects from S3 Standard to S3 Intelligent-Tiering

D. Use S3 Inventory to identify and transition objects that have not been accessed from S3 Standard to S3 Intelligent-Tiering

【參考答案】C

【題號】question 470

【題目】

A company has applications hosted on Amazon EC2 instances with IPv6 addresses. The applications must initiate communications with other external applications using the internet. However the company's security policy states that any external service cannot initiate a connection to the EC2 instances.

What should a solutions architect recommend to resolve this issue?

一家公司在具有 IPv6 位址的 Amazon EC2 執行個體上託管應用程式。應用程式必須使用網際網路啟動與其他外部應用程式的通訊。然而，該公司的安全策略規定，任何外部服務都無法啟動與 EC2 執行個體的連線。

解決方案架構師應該建議什麼來解決這個問題？

【選項】

- A.Create a NAT gateway and make it the destination of the subnet's route table
- B.Create an internet gateway and make it the destination of the subnet's route table
- C.Create a virtual private gateway and make it the destination of the subnet's route table
- D.Create an egress-only internet gateway and make it the destination of the subnet's route table

【參考答案】D

【題號】question 471

【題目】

A company is creating an application that runs on containers in a VPC. The application stores and accesses data in an Amazon S3 bucket. During the development phase, the application will store and access 1 TB of data in Amazon S3 each day. The company wants to minimize costs and wants to prevent traffic from traversing the internet whenever possible.

Which solution will meet these requirements?

一家公司正在創建一個在 VPC 中的容器上運行的應用程式。該應用程式在 Amazon S3 儲存桶中儲存和存取資料。在開發階段，該應用程式將每天在 Amazon S3 中儲存和存取 1 TB 的資料。該公司希望最大限度地降低成本，並希望盡可能阻止流量通過網路。

哪種解決方案可以滿足這些要求？

【選項】

- A.Enable S3 Intelligent-Tiering for the S3 bucket

B.Enable S3 Transfer Acceleration for the S3 bucket

C.Create a gateway VPC endpoint for Amazon S3. Associate this endpoint with all route tables in the VPC

D.Create an interface endpoint for Amazon S3 in the VPC. Associate this endpoint with all route tables in the VPC

【參考答案】C

【題號】question 472

【題目】

A company has a mobile chat application with a data store based in Amazon DynamoDB. Users would like new messages to be read with as little latency as possible. A solutions architect needs to design an optimal solution that requires minimal application changes.

Which method should the solutions architect select?

一家公司有一個行動聊天應用程式，其資料儲存是基於 Amazon DynamoDB。用戶希望以盡可能短的延遲讀取新訊息。解決方案架構師需要設計一個需要最少應用程式變更的最佳解決方案。

解決方案架構師應該選擇哪一種方法？

【選項】

A.Configure Amazon DynamoDB Accelerator (DAX) for the new messages table. Update the code to use the DAX endpoint.

B.Add DynamoDB read replicas to handle the increased read load. Update the application to point to the read endpoint for the read replicas.

C.Double the number of read capacity units for the new messages table in DynamoDB. Continue to use the existing DynamoDB endpoint.

D.Add an Amazon ElastiCache for Redis cache to the application stack. Update the application to point to the Redis cache endpoint instead of DynamoDB.

【參考答案】A

【題號】question 473

【題目】

A company hosts a website on Amazon EC2 instances behind an Application Load Balancer (ALB). The website serves static content. Website traffic is increasing, and the company is concerned about a potential increase in cost.

一家公司在應用程式負載平衡器 (ALB) 後面的 Amazon EC2 執行個體上託管一個網站。該網站提供靜態內容。網站流量不斷增加，該公司擔心成本可能會增加。

【選項】

A.Create an Amazon CloudFront distribution to cache static files at edge locations

B.Create an Amazon ElastiCache cluster. Connect the ALB to the ElastiCache cluster to serve cached files

C.Create an AWS WAF web ACL and associate it with the ALB. Add a rule to the web ACL to cache static files

D.Create a second ALB in an alternative AWS Region. Route user traffic to the closest Region to minimize data transfer costs

【參考答案】A

【題號】question 474

【題目】

A company has multiple VPCs across AWS Regions to support and run workloads that are isolated from workloads in other Regions. Because of a recent application launch requirement, the company's VPCs must communicate with all other VPCs across all Regions.

Which solution will meet these requirements with the LEAST amount of administrative effort?

一家公司在 AWS 區域擁有多個 VPC，以支援和運行與其他區域的工作負載隔離的工作負載。由於最近的應用程式啟動要求，公司的 VPC 必須與所有區域的所有其他 VPC 進行通訊。

哪一種解決方案能夠以最少的管理工作來滿足這些要求？

【選項】

A. Use VPC peering to manage VPC communication in a single Region. Use VPC peering across Regions to manage VPC communications.

B. Use AWS Direct Connect gateways across all Regions to connect VPCs across regions and manage VPC communications.

C. Use AWS Transit Gateway to manage VPC communication in a single Region and Transit Gateway peering across Regions to manage VPC communications.

D. Use AWS PrivateLink across all Regions to connect VPCs across Regions and manage VPC communications

【參考答案】C

【題號】question 475

【題目】

A company is designing a containerized application that will use Amazon Elastic Container Service (Amazon ECS). The application needs to access a shared file system that is highly durable and can recover data to another AWS Region with a recovery point objective (RPO) of 8 hours. The file system needs to provide a mount target in each Availability Zone within a Region.

A solutions architect wants to use AWS Backup to manage the replication to another Region.

Which solution will meet these requirements?

一家公司正在設計一個將使用 Amazon Elastic Container Service (Amazon ECS) 的容器化應用程式。應用程式需要存取高度耐用的共用檔案系統，並且可以將資料還原到另一個 AWS 區域，恢復點目標 (RPO) 為 8 小時。檔案系統需要為區域內的每個可用區提供掛載目標。

解決方案架構師希望使用 AWS Backup 來管理到另一個區域的複製。

哪種解決方案可以滿足這些要求？

【選項】

- A.Amazon FSx for Windows File Server with a Multi-AZ deployment
- B.Amazon FSx for NetApp ONTAP with a Multi-AZ deployment
- C.Amazon Elastic File System (Amazon EFS) with the Standard storage class
- D.Amazon FSx for OpenZFS

【參考答案】C

【題號】question 476

【題目】

A company is expecting rapid growth in the near future. A solutions architect needs to configure existing users and grant permissions to new users on AWS. The solutions architect has decided to create IAM groups. The solutions architect will add the new users to IAM groups based on department.

Which additional action is the MOST secure way to grant permissions to the new users?

一家公司預計在不久的將來會快速成長。解決方案架構師需要在 AWS 上配置現有使用者並向新使用者授予權限。解決方案架構師已決定建立 IAM 群組。解決方案架構師將根據部門將新使用者新增至 IAM 群組。

哪種附加操作是向新使用者授予權限的最安全的方式？

【選項】

- A.Apply service control policies (SCPs) to manage access permissions

- B.Create IAM roles that have least privilege permission. Attach the roles to the IAM groups
- C.Create an IAM policy that grants least privilege permission. Attach the policy to the IAM groups
- D.Create IAM roles. Associate the roles with a permissions boundary that defines the maximum permissions

【參考答案】C

【題號】question 477

【題目】

A group requires permissions to list an Amazon S3 bucket and delete objects from that bucket. An administrator has created the following IAM policy to provide access to the bucket and applied that policy to the group. The group is not able to delete objects in the bucket. The company follows least-privilege access rules.

群組需要列出 Amazon S3 儲存桶以及從該儲存桶中刪除物件的權限。管理員已建立以下 IAM 策略來提供對儲存桶的存取權限，並將該原則套用到群組。此群組無法刪除儲存桶中的物件。該公司遵循最低權限存取規則。

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Action": [
        "s3:ListBucket",
        "s3:DeleteObject"
      ],
      "Resource": [
        "arn:aws:s3:::bucket-name"
      ],
      "Effect": "Allow"
    }
  ]
}
```

Which statement should a solutions architect add to the policy to correct bucket access?

解決方案架構師應將哪條語句新增至策略中以修正儲存桶存取？

【選項】

```
"Action": [  
    "s3:*Object"  
],  
"Resource": [  
    "arn:aws:s3:::bucket-name/*"  
],  
"Effect": "Allow"
```

A.

```
"Action": [  
    "s3:*"  
],  
"Resource": [  
    "arn:aws:s3:::bucket-name/*"  
],  
"Effect": "Allow"
```

B.

```
"Action": [  
    "s3:DeleteObject"  
],  
"Resource": [  
    "arn:aws:s3:::bucket-name*"br/>],  
"Effect": "Allow"
```

C.

```
"Action": [  
    "s3:DeleteObject"  
],  
"Resource": [  
    "arn:aws:s3:::bucket-name/*"  
],  
"Effect": "Allow"
```

D.

【參考答案】D

【題號】question 478

【題目】

A law firm needs to share information with the public. The information includes hundreds of files that must be publicly readable. Modifications or deletions of the files by anyone before a designated future date are prohibited.

Which solution will meet these requirements in the MOST secure way?

律師事務所需要與公眾分享資訊。這些資訊包括數百個必須公開可讀的文件。禁止任何人在指定的未來日期之前修改或刪除檔案。

哪一種解決方案能夠以最安全的方式滿足這些要求？

【選項】

A.Upload all files to an Amazon S3 bucket that is configured for static website hosting. Grant read-only IAM permissions to any AWS principals that access the S3 bucket until the designated date.

B.Create a new Amazon S3 bucket with S3 Versioning enabled. Use S3 Object Lock with a retention period in accordance with the designated date. Configure the S3 bucket for static website hosting. Set an S3 bucket policy to allow read-only access to the objects.

C.Create a new Amazon S3 bucket with S3 Versioning enabled. Configure an event trigger to run an AWS Lambda function in case of object modification or deletion. Configure the Lambda function to replace the objects with the original versions from a private S3 bucket.

D.Upload all files to an Amazon S3 bucket that is configured for static website hosting. Select the folder that contains the files. Use S3 Object Lock with a retention period in accordance with the designated date. Grant read-only IAM permissions to any AWS principals that access the S3 bucket.

【參考答案】B

【題號】question 479

【題目】

A company is making a prototype of the infrastructure for its new website by manually provisioning the necessary infrastructure. This infrastructure includes an Auto Scaling group, an Application Load Balancer and an Amazon RDS database. After the configuration has been thoroughly validated, the company wants the capability to immediately deploy the infrastructure for development and production use in two Availability Zones in an automated fashion.

What should a solutions architect recommend to meet these requirements?

一家公司正在透過手動配置必要的基礎設施來為其新網站製作基礎設施原型。該基礎設施包括 Auto Scaling 群組、Application Load Balancer 和 Amazon RDS 資料庫。配置經過徹底驗證後，該公司希望能夠立即以自動化方式在兩個可用區中部署基礎設施以供開發和生產使用。

解決方案架構師應該建議什麼來滿足這些要求？

【選項】

- A. Use AWS Systems Manager to replicate and provision the prototype infrastructure in two Availability Zones
- B. Define the infrastructure as a template by using the prototype infrastructure as a guide. Deploy the infrastructure with AWS CloudFormation.
- C. Use AWS Config to record the inventory of resources that are used in the prototype infrastructure. Use AWS Config to deploy the prototype infrastructure into two Availability Zones.
- D. Use AWS Elastic Beanstalk and configure it to use an automated reference to the prototype infrastructure to automatically deploy new environments in two Availability Zones.

【參考答案】B

【題號】question 480

【題目】

A business application is hosted on Amazon EC2 and uses Amazon S3 for encrypted object storage. The chief information security officer has directed that no application traffic between the two services should traverse the public internet.

Which capability should the solutions architect use to meet the compliance requirements?

業務應用程式託管在 Amazon EC2 上，並使用 Amazon S3 進行加密物件儲存。首席資訊安全官指示，兩種服務之間的應用程式流量不應穿越公共互聯網。

解決方案架構師應該使用哪種功能來滿足合規性要求？

【選項】

- A.AWS Key Management Service (AWS KMS)
- B.VPC endpoint
- C.Private subnet
- D.Virtual private gateway

【參考答案】B

【題號】question 481

【題目】

A company hosts a three-tier web application in the AWS Cloud. A Multi-AZ Amazon RDS for MySQL server forms the database layer Amazon ElastiCache forms the cache layer. The company wants a caching strategy that adds or updates data in the cache when a customer adds an item to the database. The data in the cache must always match the data in the database.

Which solution will meet these requirements?

一家公司在 AWS 雲端中託管一個三層 Web 應用程式。多可用區 Amazon RDS for MySQL 伺服器構成資料庫層 Amazon ElastiCache 構成快取層。該公司需要一種快取策略，當客戶將項目新增至資料庫時，該策略可以新增或更新快取中的資料。快取中的資料必須始終與資料庫中的資料相符。

哪種解決方案可以滿足這些要求？

【選項】

- A.Implement the lazy loading caching strategy
- B.Implement the write-through caching strategy
- C.Implement the adding TTL caching strategy
- D.Implement the AWS AppConfig caching strategy

【參考答案】B

【題號】question 482

【題目】

A company wants to migrate 100 GB of historical data from an on-premises location to an Amazon S3 bucket. The company has a 100 megabits per second (Mbps) internet connection on premises. The company needs to encrypt the data in transit to the S3 bucket. The company will store new data directly in Amazon S3.

Which solution will meet these requirements with the LEAST operational overhead?

一家公司希望將 100 GB 的歷史資料從本地位置遷移到 Amazon S3 儲存桶。該公司擁有每秒 100 兆位元 (Mbps) 的內部網路連線。該公司需要對傳輸到 S3 儲存桶的資料進行加密。該公司將把新資料直接儲存在 Amazon S3 中。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

【選項】

- A.Use the s3 sync command in the AWS CLI to move the data directly to an S3 bucket
- B.Use AWS DataSync to migrate the data from the on-premises location to an S3 bucket
- C.Use AWS Snowball to move the data to an S3 bucket
- D.Set up an IPsec VPN from the on-premises location to AWS. Use the s3 cp command in the AWS CLI to move the data directly to an S3 bucket

【參考答案】B

【題號】question 483

【題目】

A company containerized a Windows job that runs on .NET 6 Framework under a Windows container. The company wants to run this job in the AWS Cloud. The job runs every 10 minutes. The job's runtime varies between 1 minute and 3 minutes.

Which solution will meet these requirements MOST cost-effectively?

一家公司對在 Windows 容器下的 .NET 6 Framework 上執行的 Windows 作業進行了容器化。該公司希望在 AWS 雲端中運行此作業。該作業每 10 分鐘運行一次。作業的運行時間在 1 分鐘到 3 分鐘之間變化。

哪一種解決方案能夠最具成本效益地滿足這些要求？

【選項】

A.Create an AWS Lambda function based on the container image of the job. Configure Amazon EventBridge to invoke the function every 10 minutes.

B.Use AWS Batch to create a job that uses AWS Fargate resources. Configure the job scheduling to run every 10 minutes.

C.Use Amazon Elastic Container Service (Amazon ECS) on AWS Fargate to run the job. Create a scheduled task based on the container image of the job to run every 10 minutes.

D.Use Amazon Elastic Container Service (Amazon ECS) on AWS Fargate to run the job. Create a standalone task based on the container image of the job. Use Windows task scheduler to run the job every 10 minutes.

【參考答案】C

【題號】question 484

【題目】

A company wants to move from many standalone AWS accounts to a consolidated, multi-account architecture. The company plans to create many new AWS accounts for different business units. The company needs to authenticate access to these AWS accounts by using a centralized corporate directory service.

Which combination of actions should a solutions architect recommend to meet these requirements? (Choose two.)

一家公司希望從許多獨立的 AWS 帳戶遷移到整合的多帳戶架構。該公司計劃為不同的業務部門創建許多新的AWS帳戶。該公司需要使用集中式企業目錄服務來驗證對這些 AWS 帳戶的存取權。

解決方案架構師應該推薦哪種操作組合來滿足這些要求？（選兩個。）

【選項】

- A.Create a new organization in AWS Organizations with all features turned on. Create the new AWS accounts in the organization.
- B.Set up an Amazon Cognito identity pool. Configure AWS IAM Identity Center (AWS Single Sign-On) to accept Amazon Cognito authentication.
- C.Configure a service control policy (SCP) to manage the AWS accounts. Add AWS IAM Identity Center (AWS Single Sign-On) to AWS Directory Service.
- D.Create a new organization in AWS Organizations. Configure the organization's authentication mechanism to use AWS Directory Service directly.
- E.Set up AWS IAM Identity Center (AWS Single Sign-On) in the organization. Configure IAM Identity Center, and integrate it with the company's corporate directory service.

【參考答案】AE

【題號】question 485

【題目】

A company is looking for a solution that can store video archives in AWS from old news footage. The company needs to minimize costs and will rarely need to restore these files. When the files are needed, they must be available in a maximum of five minutes.

What is the MOST cost-effective solution?

一家公司正在尋找一種解決方案，可以將舊新聞片段的影片檔案儲存在 AWS 中。公司需要最大限度地降低成本，並且很少需要恢復這些文件。當需要這些文件時，它們必須在最多五分鐘內可用。

最具成本效益的解決方案是什麼？

【選項】

- A.Store the video archives in Amazon S3 Glacier and use Expedited retrievals.
- B.Store the video archives in Amazon S3 Glacier and use Standard retrievals.
- C.Store the video archives in Amazon S3 Standard-Infrequent Access (S3 Standard-IA).
- D.Store the video archives in Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA).

【參考答案】A

【題號】question 486

【題目】

A company is building a three-tier application on AWS. The presentation tier will serve a static website. The logic tier is a containerized application. This application will store data in a relational database. The company wants to simplify deployment and to reduce operational costs.

Which solution will meet these requirements?

一家公司正在 AWS 上建立三層應用程式。表示層將服務於靜態網站。邏輯層是容器化應用程式。該應用程式將資料儲存在關聯式資料庫中。該公司希望簡化部署並降低營運成本。

哪種解決方案可以滿足這些要求？

【選項】

A. Use Amazon S3 to host static content. Use Amazon Elastic Container Service (Amazon ECS) with AWS Fargate for compute power. Use a managed Amazon RDS cluster for the database.

B. Use Amazon CloudFront to host static content. Use Amazon Elastic Container Service (Amazon ECS) with Amazon EC2 for compute power. Use a managed Amazon RDS cluster for the database.

C. Use Amazon S3 to host static content. Use Amazon Elastic Kubernetes Service (Amazon EKS) with AWS Fargate for compute power. Use a managed Amazon RDS cluster for the database.

D. Use Amazon EC2 Reserved Instances to host static content. Use Amazon Elastic Kubernetes Service (Amazon EKS) with Amazon EC2 for compute power. Use a managed Amazon RDS cluster for the database.

【參考答案】A

【題號】question 487

【題目】

A company seeks a storage solution for its application. The solution must be highly available and scalable. The solution also must function as a file system be mountable by multiple Linux instances in AWS and on premises through native protocols, and have no minimum size requirements. The company has set up a Site-to-Site VPN for access from its on-premises network to its VPC.

Which storage solution meets these requirements?

一家公司為其應用程式尋求儲存解決方案。該解決方案必須具有高可用性和可擴展性。此解決方案還必須充當檔案系統，可由 AWS 中和本地的多個 Linux 執行個體透過本機協定安裝，且沒有最小大小要求。該公司建立了一個站點到站點 VPN，用於從其本地網路存取其 VPC。

哪種儲存解決方案可以滿足這些要求？

【選項】

- A.Amazon FSx Multi-AZ deployments
- B.Amazon Elastic Block Store (Amazon EBS) Multi-Attach volumes
- C.Amazon Elastic File System (Amazon EFS) with multiple mount targets
- D.Amazon Elastic File System (Amazon EFS) with a single mount target and multiple access points

【參考答案】C

【題號】question 488

【題目】

A 4-year-old media company is using the AWS Organizations all features feature set to organize its AWS accounts. According to the company's finance team, the billing information on the member accounts must not be accessible to anyone, including the root user of the member accounts.

Which solution will meet these requirements?

一家成立 4 年的媒體公司正在使用 AWS Organizations 所有功能集來組織其 AWS 帳戶。據公司財務團隊稱，會員帳戶的帳單資訊不得被任何人訪問，包括會員帳戶的根用戶。

哪種解決方案可以滿足這些要求？

【選項】

- A.Add all finance team users to an IAM group. Attach an AWS managed policy named Billing to the group.
- B.Attach an identity-based policy to deny access to the billing information to all users, including the root user.
- C.Create a service control policy (SCP) to deny access to the billing information. Attach the SCP to the root organizational unit (OU).
- D.Convert from the Organizations all features feature set to the Organizations consolidated billing feature set.

【參考答案】C

【題號】question 489

【題目】

An ecommerce company runs an application in the AWS Cloud that is integrated with an on-premises warehouse solution. The company uses Amazon Simple Notification Service (Amazon SNS) to send order messages to an on-premises HTTPS endpoint so the warehouse application can process the orders. The local data center team has detected that some of the order messages were not received.

A solutions architect needs to retain messages that are not delivered and analyze the messages for up to 14 days.

Which solution will meet these requirements with the LEAST development effort?

一家電子商務公司在 AWS 雲端中運行與本地倉庫解決方案整合的應用程式。該公司使用 Amazon Simple Notification Service (Amazon SNS) 將訂單訊息傳送到本地 HTTPS 終端節點，以便倉庫應用程式可以處理訂單。本地資料中心團隊發現部分訂單訊息未收到。

解決方案架構師需要保留未傳遞的訊息並分析這些訊息最多 14 天。

哪種解決方案能夠以最少的開發工作滿足這些要求？

【選項】

A. Configure an Amazon SNS dead letter queue that has an Amazon Kinesis Data Stream target with a retention period of 14 days.

B. Add an Amazon Simple Queue Service (Amazon SQS) queue with a retention period of 14 days between the application and Amazon SNS.

C. Configure an Amazon SNS dead letter queue that has an Amazon Simple Queue Service (Amazon SQS) target with a retention period of 14 days.

D. Configure an Amazon SNS dead letter queue that has an Amazon DynamoDB target with a TTL attribute set for a retention period of 14 days.

【參考答案】C

【題號】question 490

【題目】

A gaming company uses Amazon DynamoDB to store user information such as geographic location, player data, and leaderboards. The company needs to configure continuous backups to an Amazon S3 bucket with a minimal amount of coding. The backups must not affect availability of the application and must not affect the read capacity units (RCUs) that are defined for the table.

Which solution meets these requirements?

一家遊戲公司使用 Amazon DynamoDB 來儲存用戶訊息，例如地理位置、玩家資料和排行榜。該公司需要使用最少的編碼來配置 Amazon S3 儲存桶的連續備份。備份不得影響應用程式的可用性，也不得影響為表定義的讀取容量單位 (RCU)。

哪種解決方案滿足這些要求？

【選項】

- A. Use an Amazon EMR cluster. Create an Apache Hive job to back up the data to Amazon S3.
- B. Export the data directly from DynamoDB to Amazon S3 with continuous backups. Turn on point-in-time recovery for the table.
- C. Configure Amazon DynamoDB Streams. Create an AWS Lambda function to consume the stream and export the data to an Amazon S3 bucket.
- D. Create an AWS Lambda function to export the data from the database tables to Amazon S3 on a regular basis. Turn on point-in-time recovery for the table.

【參考答案】B

【題號】question 491

【題目】

A solutions architect is designing an asynchronous application to process credit card data validation requests for a bank. The application must be secure and be able to process each request at least once.

Which solution will meet these requirements MOST cost-effectively?

解決方案架構師正在設計一個非同步應用程式來處理銀行的信用卡資料驗證請求。應用程式必須是安全的並且能夠至少處理每個請求一次。

哪一種解決方案能夠最具成本效益地滿足這些要求？

【選項】

A. Use AWS Lambda event source mapping. Set Amazon Simple Queue Service (Amazon SQS) standard queues as the event source. Use AWS Key Management Service (SSE-KMS) for encryption. Add the kms:Decrypt permission for the Lambda execution role.

B. Use AWS Lambda event source mapping. Use Amazon Simple Queue Service (Amazon SQS) FIFO queues as the event source. Use SQS managed encryption keys (SSE-SQS) for encryption. Add the encryption key invocation permission for the Lambda function.

C. Use the AWS Lambda event source mapping. Set Amazon Simple Queue Service (Amazon SQS) FIFO queues as the event source. Use AWS KMS keys (SSE-KMS). Add the kms:Decrypt permission for the Lambda execution role.

D. Use the AWS Lambda event source mapping. Set Amazon Simple Queue Service (Amazon SQS) standard queues as the event source. Use AWS KMS keys (SSE-KMS) for encryption. Add the encryption key invocation permission for the Lambda function.

【參考答案】A

【題號】question 492

【題目】

A company has multiple AWS accounts for development work. Some staff consistently use oversized Amazon EC2 instances, which causes the company to exceed the yearly budget for the development accounts. The company wants to centrally restrict the creation of AWS resources in these accounts.

Which solution will meet these requirements with the LEAST development effort?

一家公司擁有多個用於開發工作的 AWS 帳戶。一些員工持續使用超大的 Amazon EC2 實例，這導致公司超出了開發帳戶的年度預算。該公司希望集中限制這些帳戶中 AWS 資源的建立。

哪種解決方案能夠以最少的開發工作滿足這些要求？

【選項】

- A. Develop AWS Systems Manager templates that use an approved EC2 creation process. Use the approved Systems Manager templates to provision EC2 instances.
- B. Use AWS Organizations to organize the accounts into organizational units (OUs). Define and attach a service control policy (SCP) to control the usage of EC2 instance types.
- C. Configure an Amazon EventBridge rule that invokes an AWS Lambda function when an EC2 instance is created. Stop disallowed EC2 instance types.
- D. Set up AWS Service Catalog products for the staff to create the allowed EC2 instance types. Ensure that staff can deploy EC2 instances only by using the Service Catalog products.

【參考答案】B

【題號】question 493

【題目】

A company wants to use artificial intelligence (AI) to determine the quality of its customer service calls. The company currently manages calls in four different languages, including English. The company will offer new languages in the future. The company does not have the resources to regularly maintain machine learning (ML) models.

The company needs to create written sentiment analysis reports from the customer service call recordings. The customer service call recording text must be translated into English.

Which combination of steps will meet these requirements? (Choose three.)

一家公司希望使用人工智慧 (AI) 來確定其客戶服務呼叫的品質。該公司目前管理四種不同語言的通話，其中包括英語。該公司未來將提供新的語言。該公司沒有資源來定期維護機器學習 (ML) 模型。

該公司需要根據客戶服務通話錄音建立書面情緒分析報告。客戶服務通話錄音文字必須翻譯成英文。

哪一種步驟組合可以滿足這些要求？（選擇三項。）

【選項】

- A. Use Amazon Comprehend to translate the audio recordings into English.
- B. Use Amazon Lex to create the written sentiment analysis reports.
- C. Use Amazon Polly to convert the audio recordings into text.
- D. Use Amazon Transcribe to convert the audio recordings in any language into text.
- E. Use Amazon Translate to translate text in any language to English.
- F. Use Amazon Comprehend to create the sentiment analysis reports.

【參考答案】DEF

【題號】question 494

【題目】

A company uses Amazon EC2 instances to host its internal systems. As part of a deployment operation, an administrator tries to use the AWS CLI to terminate an EC2 instance. However, the administrator receives a 403 (Access Denied) error message.

The administrator is using an IAM role that has the following IAM policy attached:

What is the cause of the unsuccessful request?

一家公司使用 Amazon EC2 執行個體來託管其內部系統。作為部署作業的一部分，管理員嘗試使用 AWS CLI 終止 EC2 執行個體。但是，管理員收到 403(訪問被拒絕) 錯誤訊息。

管理者正在使用附加了以下 IAM 策略的 IAM 角色：

請求失敗的原因是什麼？

```

{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": ["ec2:TerminateInstances"],
      "Resource": ["*"]
    },
    {
      "Effect": "Deny",
      "Action": ["ec2:TerminateInstances"],
      "Condition": {
        "NotIpAddress": {
          "aws:SourceIp": [
            "192.0.2.0/24",
            "203.0.113.0/24"
          ]
        }
      }
    }
  ],
  "Resource": ["*"]
}

```

【選項】

- A.The EC2 instance has a resource-based policy with a Deny statement.
- B.The principal has not been specified in the policy statement.
- C.The "Action" field does not grant the actions that are required to terminate the EC2 instance.
- D.The request to terminate the EC2 instance does not originate from the CIDR blocks 192.0.2.0/24 or 203.0.113.0/24.

【參考答案】D

【題號】question 495

【題目】

A company is conducting an internal audit. The company wants to ensure that the data in an Amazon S3 bucket that is associated with the company's AWS Lake Formation data lake does not contain sensitive customer or employee data. The company wants to discover personally identifiable information (PII) or financial information, including passport numbers and credit card numbers.

Which solution will meet these requirements?

某公司正在進行內部稽核。該公司希望確保與該公司的 AWS Lake Formation 資料湖關聯的 Amazon S3 儲存桶中的資料不包含敏感的客戶或員工資料。該公司希望發現個人識別資訊 (PII) 或財務信息，包括護照號碼和信用卡號碼。

哪種解決方案可以滿足這些要求？

【選項】

A. Configure AWS Audit Manager on the account. Select the Payment Card Industry Data Security Standards (PCI DSS) for auditing.

B. Configure Amazon S3 Inventory on the S3 bucket. Configure Amazon Athena to query the inventory.

C. Configure Amazon Macie to run a data discovery job that uses managed identifiers for the required data types.

D. Use Amazon S3 Select to run a report across the S3 bucket.

【參考答案】C

【題號】question 496

【題目】

A company uses on-premises servers to host its applications. The company is running out of storage capacity. The applications use both block storage and NFS storage. The

company needs a high-performing solution that supports local caching without re-architecting its existing applications.

Which combination of actions should a solutions architect take to meet these requirements? (Choose two.)

一家公司使用本地伺服器來託管其應用程式。該公司的存儲容量即將耗盡。這些應用程式同時使用區塊儲存和 NFS 儲存。該公司需要一個高效能的解決方案，支援本地緩存，而無需重新建置現有應用程式。

解決方案架構師應該採取哪些操作組合來滿足這些要求？（選兩個。）

【選項】

- A. Mount Amazon S3 as a file system to the on-premises servers.
- B. Deploy an AWS Storage Gateway file gateway to replace NFS storage.
- C. Deploy AWS Snowball Edge to provision NFS mounts to on-premises servers.
- D. Deploy an AWS Storage Gateway volume gateway to replace the block storage.
- E. Deploy Amazon Elastic File System (Amazon EFS) volumes and mount them to on-premises servers.

【參考答案】BD

【題號】question 497

【題目】

A company has a service that reads and writes large amounts of data from an Amazon S3 bucket in the same AWS Region. The service is deployed on Amazon EC2 instances within the private subnet of a VPC. The service communicates with Amazon S3 over a NAT gateway in the public subnet. However, the company wants a solution that will reduce the data output costs.

Which solution will meet these requirements MOST cost-effectively?

一家公司有一項服務，可以從同一 AWS 區域中的 Amazon S3 儲存桶讀取和寫入大量資料。該服務部署在 VPC 私有子網路內的 Amazon EC2 執行個體上。本服務透過公有子網路中的

NAT 閘道與 Amazon S3 進行通訊。然而，該公司想要一種能夠降低資料輸出成本的解決方案。

哪一種解決方案能夠最具成本效益地滿足這些要求？

【選項】

A.Provision a dedicated EC2 NAT instance in the public subnet. Configure the route table for the private subnet to use the elastic network interface of this instance as the destination for all S3 traffic.

B.Provision a dedicated EC2 NAT instance in the private subnet. Configure the route table for the public subnet to use the elastic network interface of this instance as the destination for all S3 traffic.

C.Provision a VPC gateway endpoint. Configure the route table for the private subnet to use the gateway endpoint as the route for all S3 traffic.

D.Provision a second NAT gateway. Configure the route table for the private subnet to use this NAT gateway as the destination for all S3 traffic.

【參考答案】C

【題號】question 498

【題目】

A company uses Amazon S3 to store high-resolution pictures in an S3 bucket. To minimize application changes, the company stores the pictures as the latest version of an S3 object. The company needs to retain only the two most recent versions of the pictures.

The company wants to reduce costs. The company has identified the S3 bucket as a large expense.

Which solution will reduce the S3 costs with the LEAST operational overhead?

一家公司使用 Amazon S3 將高解析度圖片儲存在 S3 儲存桶中。為了最大限度地減少應用程式更改，該公司將圖片儲存為 S3 物件的最新版本。該公司只需保留圖片的兩個最新版本。

該公司希望降低成本。該公司已將 S3 儲存桶確定為一項重大支出。

哪種解決方案能夠以最少的營運開銷降低 S3 成本？

【選項】

- A. Use S3 Lifecycle to delete expired object versions and retain the two most recent versions.
- B. Use an AWS Lambda function to check for older versions and delete all but the two most recent versions.
- C. Use S3 Batch Operations to delete noncurrent object versions and retain only the two most recent versions.
- D. Deactivate versioning on the S3 bucket and retain the two most recent versions.

【參考答案】A

【題號】question 499

【題目】

A company needs to minimize the cost of its 1 Gbps AWS Direct Connect connection. The company's average connection utilization is less than 10%. A solutions architect must recommend a solution that will reduce the cost without compromising security.

Which solution will meet these requirements?

公司需要最大限度地降低 1 Gbps AWS Direct Connect 連接的成本。公司平均連線利用率不足 10%。解決方案架構師必須推薦一個既能降低成本又不會影響安全性的解決方案。

哪種解決方案可以滿足這些要求？

【選項】

- A. Set up a new 1 Gbps Direct Connect connection. Share the connection with another AWS account.
- B. Set up a new 200 Mbps Direct Connect connection in the AWS Management Console.

C.Contact an AWS Direct Connect Partner to order a 1 Gbps connection. Share the connection with another AWS account.

D.Contact an AWS Direct Connect Partner to order a 200 Mbps hosted connection for an existing AWS account.

【參考答案】D

【題號】question 500

【題目】

A company has multiple Windows file servers on premises. The company wants to migrate and consolidate its files into an Amazon FSx for Windows File Server file system. File permissions must be preserved to ensure that access rights do not change.

Which solutions will meet these requirements? (Choose two.)

一家公司擁有多個本機 Windows 檔案伺服器。該公司希望將其檔案遷移並整合到 Amazon FSx for Windows File Server 檔案系統中。必須保留檔案權限以確保存取權限不會變更。

哪些解決方案可以滿足這些要求？（選兩個。）

【選項】

A.Deploy AWS DataSync agents on premises. Schedule DataSync tasks to transfer the data to the FSx for Windows File Server file system.

B.Copy the shares on each file server into Amazon S3 buckets by using the AWS CLI. Schedule AWS DataSync tasks to transfer the data to the FSx for Windows File Server file system.

C.Remove the drives from each file server. Ship the drives to AWS for import into Amazon S3. Schedule AWS DataSync tasks to transfer the data to the FSx for Windows File Server file system.

D.Order an AWS Snowcone device. Connect the device to the on-premises network. Launch AWS DataSync agents on the device. Schedule DataSync tasks to transfer the data to the FSx for Windows File Server file system.

E. Order an AWS Snowball Edge Storage Optimized device. Connect the device to the on-premises network. Copy data to the device by using the AWS CLI. Ship the device back to AWS for import into Amazon S3. Schedule AWS DataSync tasks to transfer the data to the FSx for Windows File Server file system.

【參考答案】AD

領先制勝